
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

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ABSTRACT

Uniform manifold approximation and projection (UMAP) is a leading neighbor-embedding method, yet the forces that produce its visual structure remain under-characterized. We analyze its attractive and repulsive forces directly and show how their functional shapes determine cluster formation, visualization quality, and the differences between UMAP and related methods. Repulsion principally controls cluster boundaries and inter-cluster distances. Attraction is the decisive and more subtle mechanism: the same attractive update can contract or expand low-dimensional distances, which explains UMAP’s dependence on learning-rate annealing and clarifies why attractive and repulsive terms must be treated differently. We further show that a direct modification of the attraction shape improves the consistency of cluster formation under random initialization. The resulting framework gives a sharper interpretation of UMAP and related embedding algorithms and provides concrete routes to more robust and accurate embeddings.

1 Introduction

Modern data analysis is dominated by high-dimensional measurements whose structure is difficult to inspect directly. Dimensionality reduction (DR) techniques therefore serve as essential tools for exploratory analysis because they expose this structure through low-dimensional visualization. The most widely used neighbor-embedding methods, t -distributed stochastic neighbor embedding (Maaten and Hinton [2008]) and uniform manifold approximation and projection (UMAP) (McInnes et al. [2018]), are driven by attraction-repulsion dynamics that pull similar data points together and push dissimilar ones apart. As unsupervised algorithms, they do not require labeled data; instead, they recover and preserve intrinsic high-dimensional structure by balancing local attractive relationships with global repulsive ones. This capability makes them especially valuable for clustering (Becht et al. [2019]), exploratory data analysis (Fleischer and Islam [2020]), anomaly detection in semiconductor manufacturing (Fan et al. [2021]), visual search (González-Márquez et al. [2024]), time series analysis (Altin and Cakir [2024]), studying representation convergence (Huh et al. [2024]), and outlier image detection (Islam and Fleischer [2024]), where hidden patterns in unlabeled data must be made visible. DR thus demonstrates the practical power of unsupervised learning: it converts complex data into interpretable geometric structure without supervision.

Newer methods build on the same attraction-repulsion principle (Amid and Warmuth [2019], Agrawal et al. [2021], Wang et al. [2021], Narayan et al. [2021], Yang et al. [2022], Wang et al. [2025], Kury et al. [2025]), but tune that principle to emphasize different aspects of the data. Despite their broad use, these methods still rely on heuristics that can obscure the meaning of the resulting embeddings. Because DR necessarily introduces distortions (Chari and Pachter [2023]), reliable interpretation requires a mechanistic account of how the algorithms create visible structure, avoid spurious patterns, and respond to parameter choices. In practice, compact clusters are obtained through initialization choices, learning-rate schedules, and kernel tuning, yet the attractive and repulsive dynamics that actually create those clusters have remained insufficiently resolved. The central tunable quantities are embedded inside abstract functional forms, which makes the algorithms harder to explain precisely.

This paper resolves that gap by decomposing the forces into their constituent parts and extracting the functional shapes of attraction and repulsion. We show that attraction, not repulsion, is the central mechanism behind learning-rate annealing, consistency under random initialization, and the formation of clusters. Repulsion plays the complementary and more specific role of regulating inter-cluster distances. Our specific contributions are:

1. We formulate attraction and repulsion shapes directly from the corresponding forces, derive explicit conditions for distance contraction and expansion, and establish a mechanistic lens for these algorithms (Section 4).
2. We show that UMAP’s attraction shape produces both contraction and expansion of distance, a counterintuitive behavior that decisively explains the role of learning-rate annealing (Section 5).
3. We modify the attraction shape and obtain more consistent embeddings under random initialization, supporting the conclusion that UMAP encodes a reproducible underlying structure (Section 5.1).
4. We analyze repulsion shapes to separate the mechanisms of cluster formation from the mechanisms that regulate inter-cluster distance (Section 5.2).
5. We compare the attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG-*t*-SNE, identify their precise similarities and differences, and characterize their stability (Section 6).

We focus the main text on UMAP because it has become a de facto standard across fields, and we use other algorithms to sharpen the comparisons when they clarify the force-shape mechanism. Additional algorithmic details are provided in the appendix. The codes used in the research will be released upon acceptance.

2 Related Works

Modern iterative graph-based neighbor embedding begins with stochastic neighbor embedding (SNE) (Hinton and Roweis 2002) and its *t*-distributed extension (*t*-SNE) (Maaten and Hinton, 2008). These methods use dense graphs in which every point has a pairwise relation with every other point, independent of whether the pair is similar. Their graph weights are normalized to define probability distributions. In parallel, methods such as locally linear embedding (Roweis and Saul, 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi 2002) used *k*-nearest neighbor (*k*-NN) graphs to encode pairwise interactions. These spectral methods rely on Eigenvalue decomposition. Tang et al. (Tang et al., 2016) then brought the *k*-NN graph into an iterative optimization framework and removed low-dimensional normalization. UMAP (McInnes et al., 2018) completed this move by removing normalization in both high and low dimensions and constructing the embedding from pairwise interactions alone. Its optimization explicitly uses attraction to preserve local neighborhoods and repulsion to separate dissimilar points. PaCMAP (Wang et al., 2021) and NEG-*t*-SNE (Damrich et al., 2023) build directly on this foundation, and a recent survey is provided by de Bodt et al. (2025).

Substantial progress has clarified relationships among these algorithms, but the force-level mechanisms still require a sharper account. Early analysis of SNE showed that well-clustered original data remain well-clustered in the embedding space (Shaham and Steinerberger, 2017). For *t*-SNE, Linderman and Steinerberger (2019) proved that the number of embedded clusters lower-bounds the number of clusters in the original space, with subsequent work providing further characterization (Arora et al., 2018; Cai and Ma 2022; Linderman and Steinerberger 2022). Because *t*-SNE and UMAP share a common conceptual origin while producing markedly different visualizations, connecting them has become a central line of work (Böhm et al., 2022; Damrich et al., 2023; Draganov et al., 2023). Böhm et al. (2022) proposed that Laplacian Eigenmap, ForceAtlas2 (Jacomy et al., 2014), UMAP, and *t*-SNE sample a common underlying spectrum. LE and *t*-SNE are indeed linked through the early exaggeration phase (Cai and Ma 2022), and UMAP and *t*-SNE can be related through contrastive estimation (Damrich et al., 2023). Around the same time, Hu et al. (2023) independently identified the connection between contrastive learning and SNE. Recent work further explains DR through probabilistic formulations (Ravuri et al., 2023; Ravuri and Lawrence 2024), kernel techniques (Draganov and Dohn, 2023), and information geometry (Kolpakov and Rocke, 2024).

3 Uniform Manifold Approximation and Projection

UMAP begins by constructing a high-dimensional graph for the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ through the pairwise relation $p_{i,j} = f_h(d(x_i, x_j))$, typically with values in $[0, 1]$ (for details, see Appendix H).

The low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is then encoded by the differentiable affinity function

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

where the parameters a and b control the density of the mapping and are selected by fitting q_{ij} to

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d, \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

where m_d regulates the minimum distance between nearest low-dimensional points. UMAP minimizes the cross-entropy loss

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

The first term generates the attractive force, and the second term generates the repulsive force. Rather than optimizing all pairs at every iteration, UMAP uses negative sampling (Mikolov et al. 2013, Tang et al. 2016): for each positive edge with $p_{ij} > 0$, it samples several negative edges at random. The attractive force updates the positive edge as

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

and the repulsive force updates the negative edges as

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ is the learning rate and t is the step number. For negative edges, y_j is not updated. A detailed analysis of UMAP's loss is given by (Damrich and Hamprecht 2021).

4 Attraction and Repulsion Shapes

The updates (4)-(6) become interpretable once the gradients $\nabla_{y_i^t} \log(q_{ij})$ ($\nabla_{y_i^t} \log(1 - q_{ij})$) are decomposed into a distance-dependent scalar coefficient multiplying the vector $(y_i - y_j)$. We call this scalar coefficient the attraction (repulsion) shape. Although UMAP is our primary example, the formalism applies directly to any embedding method built from attraction and repulsion.

Writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$, where $\zeta = \|y_i - y_j\|_2$, gives the positive-edge update

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here, $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ is the attraction shape, and the Euclidean metric gives $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$. The same decomposition yields the negative-edge update

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape.

Related decompositions appear in prior work (McInnes et al. 2018, Agrawal et al. 2021, Draganov and Dohn 2023), but their roles differ from ours. The original UMAP paper (McInnes et al. 2018) used the decomposition mainly as a computational device, and (Draganov and Dohn 2023) used it to compare algorithms through kernels; in both cases, the derivative itself was the main object. We instead elevate the decomposed terms to independent functions whose shapes determine embedding behavior. (Agrawal et al. 2021) used a related decomposition into a scalar coefficient and a unit vector $((y_i - y_j)/\|y_i - y_j\|_2)$ to emphasize force magnitude. Our analysis shows that the full shape of the scalar function is more diagnostic than magnitude alone.

4.1 Conditions for Attraction and Repulsion

We now derive exact per-step conditions for attraction and repulsion from the update equations (7)-(10). The following proposition characterizes when a positive edge contracts the distance between y_i^t and y_j^t :

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

Thus, λf_a is the effective attraction shape. In most settings, however, f_a alone already gives decisive information about the embedding. For a negative edge, the following proposition characterizes when the update expands the distance between y_i^t and y_j^t :

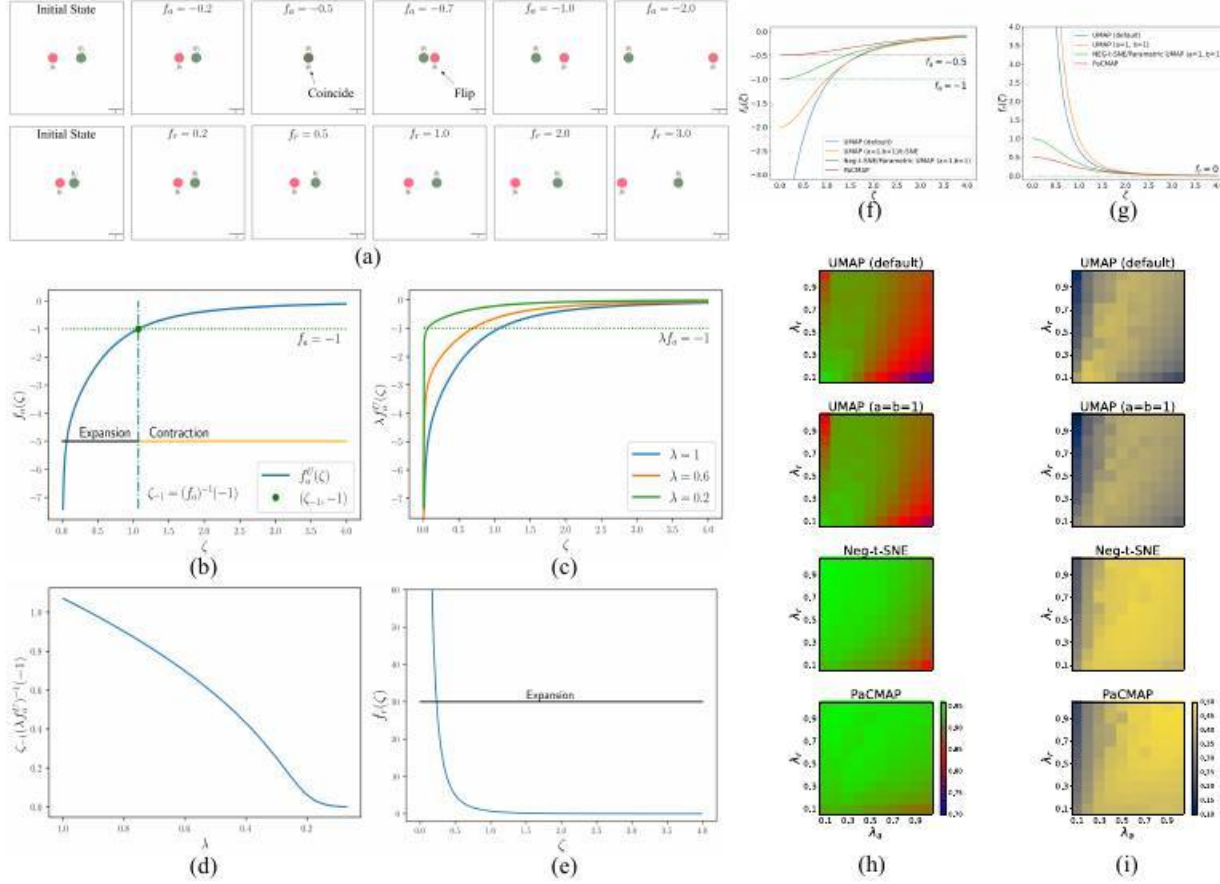


Figure 1: Attraction and repulsion shapes expose the operating mechanism of UMAP. (a) Different values of f_a (top) and f_r (bottom) determine pairwise motion. (b) UMAP’s attraction shape. (c) Effective attraction shape (λf_a) for different learning rates λ . (d) Minimum contraction distance (ζ_{-1}) as λ decreases. (e) UMAP’s repulsion shape. (f) Attraction and (g) repulsion shapes for several embedding methods. (h) Trustworthiness and (i) Silhouette Score as the constant learning rate is varied independently for attraction (λ_a) and repulsion (λ_r). Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

Proposition 4.2. The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.

Including a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ in Eq. (10) leaves this conclusion unchanged. These conditions provide a per-iteration certificate for guaranteed contraction or expansion; they are distinct from learning-rate tuning or decay because the shapes themselves can be designed so that contraction and expansion do not depend on annealing.

Figure 1(a) makes the distinction clear for two points with $f_a < 0$ and $f_r > 0$. Repulsion is direct: any positive value increases the distance. Attraction is subtler and more consequential because f_a can encode both contraction and expansion. For $f_a \in (-1.0, 0)$, the distance decreases, with a sign flip at $f_a = -0.5$, where attraction brings the points to coincidence. Values below -1.0 increase the distance instead. These local rules collectively determine the global structure of the embedding.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Combining the gradient decomposition with the distance form in (1) gives the UMAP attraction and repulsion shapes

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

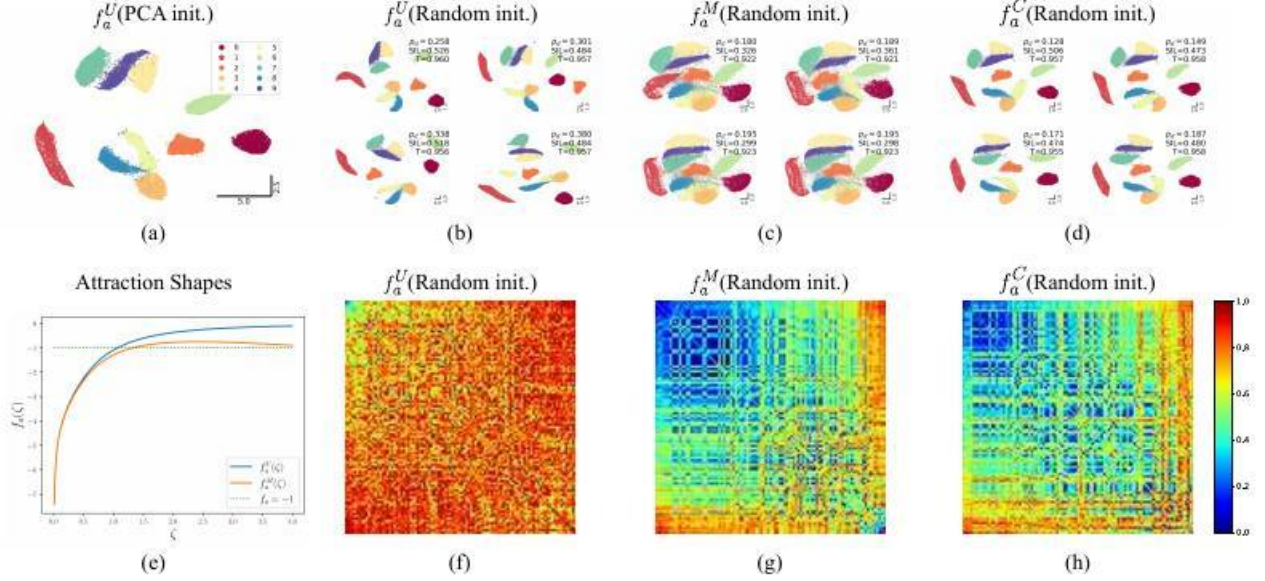


Figure 2: Random initialization reveals the decisive role of attraction shape on the MNIST dataset. (a) PCA-initialized mapping. (b-d) Four mappings with the lowest Procrustes distance (p_d) from (a) for (b) UMAP, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs of (f) UMAP (0.78 ± 0.13), (g) modified (0.49 ± 0.21), and (h) composite attraction (0.50 ± 0.20) shapes. Diagonal (i, i) entries are sorted by Procrustes distance (p_d) from (a), and off-diagonal (i, j) entries give p_d between the i^{th} and j^{th} mappings. The matrices and (mean $p_d \pm \text{std}$) values show that default UMAP embeddings are not self-similar, whereas the modified and composite attraction shapes enforce more scale-invariant structure.

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. This section analyzes the default UMAP shapes, which are controlled primarily by a and b , and uses controlled perturbations to identify their roles. The conclusions below hold generally for $b \leq 1$, where the attraction shape is strictly increasing and therefore invertible (For derivation, see Appendix A).

Figure 1(b) shows UMAP’s default attraction shape ($a = 1.58, b = 0.89$). Its magnitude diverges as $\zeta \rightarrow 0$. Proposition 4.1 therefore pinpoints the contraction-to-expansion transition: it occurs when λf_a^U crosses -1 as ζ approaches 0. Because the attraction shape is invertible, the transition distance $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$ is the minimum distance at which attractive updates still contract a pair. Thus, $\zeta > \zeta_{-1}$ yields contraction, whereas $0 < \zeta < \zeta_{-1}$ yields expansion. This directly overturns the common intuition that attractive updates always bring neighboring points closer together.

When ζ_{-1} is large, neighboring points alternate between contraction and expansion, producing fuzzy clusters. Sharp boundaries therefore require the optimization to drive the system toward $\zeta_{-1} \rightarrow 0$. This is precisely why UMAP needs its learning rate schedule to anneal toward zero (Figs. 1(c,d)). In contrast, the repulsion shape (Eq. 12) is always positive and therefore satisfies Proposition 4.2 f_r^U approaches 0 as $\zeta \rightarrow \infty$ and diverges as $\zeta \rightarrow 0$ (Fig. 1(e)).

Comparing attraction shapes across algorithms shows that UMAP and t-SNE are the methods that confront $f_a < -1$ (and therefore $\zeta_1 > 0$ at $\lambda = 1$, Fig. 1(f)). t-SNE controls this issue by weighting updates with the corresponding p_{ij} values, whereas UMAP controls it through learning-rate annealing. NEG-t-SNE and PaCMAP keep f_a naturally within $[-1, 0]$, satisfying Proposition 4.1 for $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$. PaCMAP’s weighted f_a (< 0.5 always) goes further and prevents flips during attraction. The corresponding repulsion shapes show that UMAP uniquely faces very large values at short distances (Fig. 1(g); the shape is unbounded, though optimization often clips the values). Appendix G gives additional details for these and related methods.

We isolate these attraction/repulsion effects on MNIST by using separate constant learning rates for attraction (λ_a) and repulsion (λ_r) and varying them independently (Figs. 1(h,i)). For UMAP, smaller λ_a values (< 0.5 and consequently $\zeta_{-1} \simeq 0$) give better embeddings, measured by trustworthiness (T) (Venna and Kaski 2001) in Fig. 1(h), and better clustering, measured by silhouette score (SIL) (Rousseeuw 1987) in Fig. 1(i). By contrast, the full range of $\lambda_r \in [0, 1]$

remains usable. For NEG- t -SNE and PaCMAP, where $\zeta_{-1} = 0$, the full parameter range remains effective. These results confirm that UMAP achieves satisfactory embedding and clustering by forcing ζ_{-1} close to 0 through learning-rate annealing, whereas newer algorithms build $\zeta_{-1} = 0$ into the default attraction shape. Further discussion, results, and dataset illustrations appear in Section 6 and Appendices B, G, and I.

5.1 Improving UMAP’s Consistency under Random Initialization

UMAP’s consistency depends strongly on initialization (Kobak and Linderman, 2021; Wang et al., 2021). In standard practice, principal component analysis (PCA) or spectral decomposition of the high-dimensional graph initializes the embedding and yields consistent mappings despite stochasticity. With random initialization, however, clusters often fail to form or appear in a different orientation at each run. The force-shape explanation is direct: when two high-dimensional nearest neighbors start far apart in the embedding, attraction can be too weak to contract them effectively. This near-sightedness (Wang et al., 2021) is visible in the attraction shape because $|f_a^U|$ decays toward zero with distance (i.e., $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)|\zeta = 0$).

We therefore induce “far-sightedness” by strengthening attraction at large distances so that faraway neighbors can contract. To test this mechanism, we modify UMAP’s attraction shape as

$$f_a^M = f_a^U - \beta\zeta, \quad (13)$$

where β controls the strength of the added term (we used $\beta = 0.2$, Fig. 2(e)). This attraction-shape change corresponds to adding a regularizer to the attractive term of the loss function (i.e., $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^3$). Beyond pulling distant pairs together, the modification promotes point intermixing that improves convergence from random initialization, analogous in effect to early exaggeration in t -SNE. Equation (13) uses the simplest linear correction; functions such as $\log \zeta$ or ζ^p ($p \in \mathbb{R}_{\geq 0}$) could serve the same role.

We also introduce a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases}. \quad (14)$$

The composite shape uses the modified attraction only during the early phase and then returns to the original UMAP shape to reduce distortions introduced by f_a^M . We next evaluate how the modified and composite shapes affect DR from random initialization.

We first compute a PCA-initialized MNIST embedding (Figure 2(a)). We then generate Gaussian-randomly initialized embeddings for each attraction shape and repeat the experiment 100 times. To quantify consistency, we use Procrustes analysis (Gower, 1975), which aligns two point clouds under scaling, translation, rotation, and reflection (for details, see Appendix D.1). We align each randomly initialized embedding to the PCA-initialized reference and measure the separation with Procrustes distance (p_d). Figure 2(b) shows the four UMAP embeddings with the lowest p_d ; their cluster shapes are consistent, but their placements are not. The modified and composite attraction shapes (Figs. 2(c) and (d)) produce visibly more consistent cluster placements.

We further summarize placement consistency with a Procrustes matrix: diagonal entries are sorted by p_d from the PCA-initialized mapping, and off-diagonal entries give p_d between two randomly initialized mappings (for details, see Appendix D.1). This construction parallels a similarity matrix (Foote, 1999). Default UMAP embeddings are not mutually similar (Fig. 2(f)), whereas the modified (Fig. 2(g)) and composite (Fig. 2(h)) shapes show strong mutual similarity. This supports the stronger conclusion that UMAP encodes a unique structure independent of initialization, but default attraction can fail to recover it because long-distance attraction is too weak.

5.2 Cluster Formation and Compactness

The main user-facing parameter for cluster formation in UMAP is the minimum distance m_d (through Eq. 2). However, changing m_d constrains the accessible values of a and b (Fig. 3(a)). Consequently, reducing m_d often yields embeddings with little additional benefit (Figs. 3(b-d)). The reason is clear in the shape view: varying m_d only weakly changes the repulsion shape (Fig. 3(e)). A more direct intervention is to vary a and b themselves. Decreasing a strengthens repulsion but weakens attraction even faster, worsening near-sightedness. Decreasing b provides stronger control (Figs. 3(e,i)). Figures 3(f-h) show that setting b to 0.6, 0.5, and 0.4 progressively increases inter-cluster distance and splits clusters that were previously merged. This splitting follows from the increasingly heavy-tailed kernel induced by smaller b values (heavy-tailed kernels produce smaller and more distinct clusters (Van der Maaten and Hinton, 2008; Kobak et al., 2019); see Kobak et al. (2019) for how varying b in UMAP modulates the tail). Although this approach separates all ten MNIST labels into distinct clusters, it also changes attraction, so the independent contributions of attraction and repulsion remain entangled.

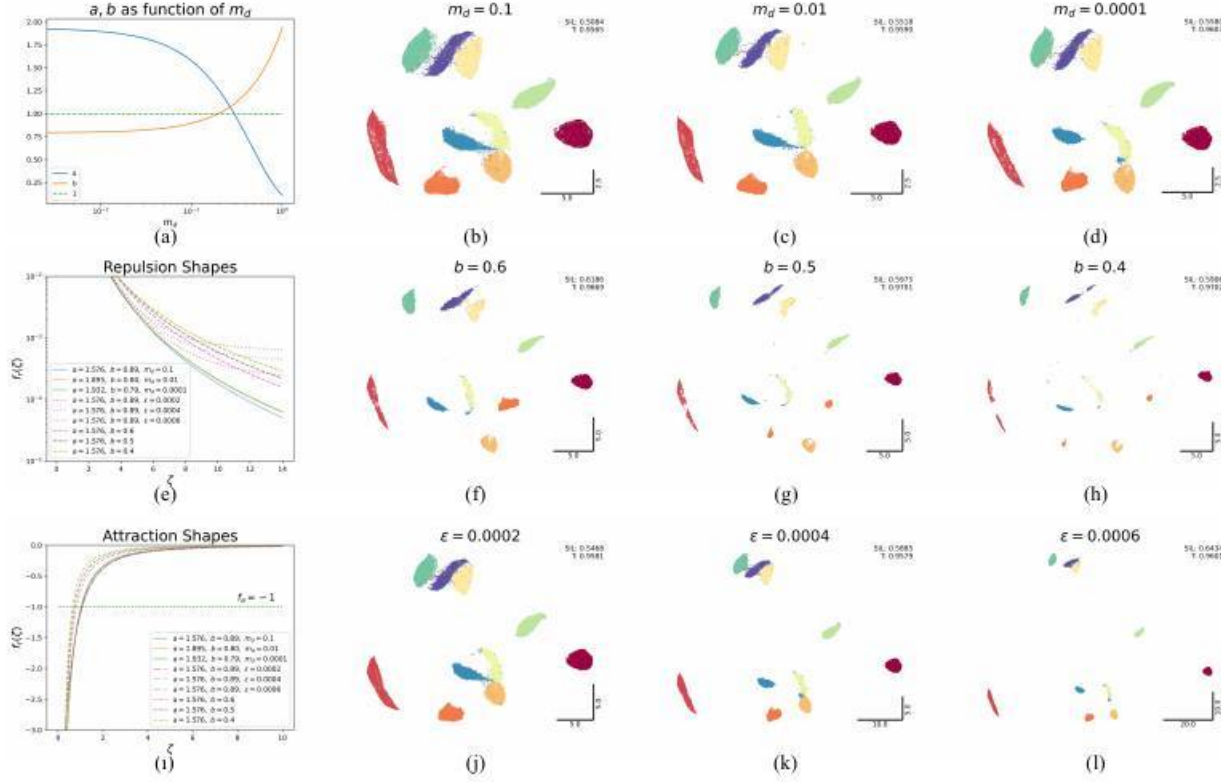


Figure 3: Repulsion controls inter-cluster distances on the MNIST dataset. (a) Computing a, b by varying the low-dimensional distance m_d restricts the available parameter exploration. (b-d) UMAP outputs for $m_d = 0.1, 0.01$, and 0.0001 show limited gains in cluster compactness. (e) Repulsion shapes for different parameters. (f-h) Explicitly increasing repulsion by varying b produces more compact clusters and reveals clusters that were otherwise absent. (i) Attraction shapes under parameter variation. (j-l) Adding a small positive value (ϵ) to the repulsion shape increases inter-cluster distance.

To isolate repulsion, we keep attraction fixed and modify only the repulsion shape by adding a small positive value (ϵ):

$$f_r^M = f_r^U + \epsilon, \quad (15)$$

while holding a and b fixed. This modification adds a regularizer to the repulsive term of the loss function (i.e., $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \epsilon/2 \|y_i - y_j\|_2^2$) and produces stronger repulsion than the default shape (Fig. 3(e)). Figs. 3(j-l) show a direct increase in inter-cluster distances as ϵ grows. The clustering behavior nevertheless resembles the behavior obtained by varying m_d , with looser separation of labels as ϵ increases. Because ϵ leaves the attraction shape unchanged, and because varying m_d traces similar attraction shapes (Fig. 3(i)), these experiments identify attraction as the dominant driver of cluster formation.

We resolve the cluster-formation mechanism by changing only one shape at a time, using $b = 0.4$ for either attraction or repulsion while keeping the other shape at its default (Fig. 4). Default attraction combined with stronger repulsion does not reveal new structures or clusters; it mainly produces smaller, more compact versions of the original UMAP clusters (Fig. 4(a)). In contrast, stronger attraction with default repulsion creates additional structures within clusters (Fig. 4(b)). Several original clusters split into smaller clusters. This establishes the division of labor: attraction creates cluster structure, while repulsion controls compactness, either by shrinking clusters or increasing inter-cluster distance depending on the repulsion shape. Appendix G.3 shows that LocalMAP embeddings exploit the same attraction-repulsion interplay to separate clusters.

6 Comparison to NEG- t -SNE

Many dimensionality reduction algorithms use the setting $a = 1$ and $b = 1$. This choice stabilizes or bounds gradients in several algorithms and thereby simplifies optimization. Damrich et al. (2023) examined this setting for UMAP and

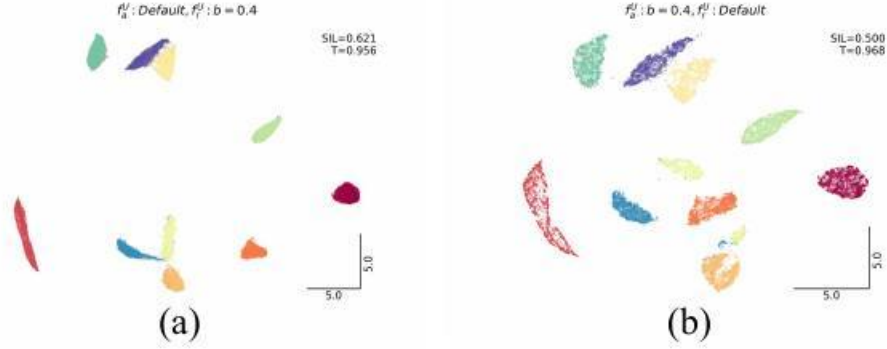


Figure 4: Embedding of the MNIST dataset after independently changing attraction and repulsion. (a) Default attraction with the repulsion shape from $b = 0.4$ preserves the default UMAP clusters while increasing compactness. (b) Default repulsion with the attraction shape from $b = 0.4$ creates new within-cluster structure and additional clusters.

proposed Neg- t -SNE, which combines stable gradients with more compact clustering even at a constant learning rate of 1. Parametric UMAP (Sainburg et al. 2021) initially used the UMAP loss, but later it adopted a numerically stable modified cross-entropy loss (Shi et al. 2023) with a logsigmoid kernel. Analyzing the two formulations shows that they reach the same objective from different starting points, yielding the following observation:

Proposition 6.1. *Neg- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

For UMAP with $a = 1$ and $b = 1$, the attraction and repulsion shapes become $f_a^U = -2/(1 + \zeta^2)$ and $f_r^U = 2/(\zeta^2(1 + \zeta^2))$, respectively. The attraction shape is bounded within $[-2, 0]$, with $f_a^U(0) = -2$ (Fig. 1(f)), but the repulsion shape remains essentially unchanged and unbounded as $\zeta \rightarrow 0$ (Fig. 1(g)). Because $f_a^U < -1$ near $\zeta = 0$, Proposition 4.1 and Section 5 imply that this unity case still requires learning-rate annealing.

Damrich et al. (2023) analyzed UMAP’s negative-sampling loss through contrastive embedding (CE) and showed that UMAP’s effective kernel is $1/\zeta^2$. Within the CE framework, NEG- t -SNE replaces this kernel with $1/(1 + \zeta^2)$. Translating that formulation back into UMAP notation gives the low-dimensional affinity $q_{ij}^N = 1/(2 + \zeta^2)$ and therefore the attraction and repulsion shapes $f_a^N = -2/(2 + \zeta^2)$, and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$, respectively (Figs. 1(f,g)). The attraction shape is bounded within $[-1, 0]$ and satisfies Proposition 4.1. Every $\lambda \in [0, 1]$ therefore contracts distances without oscillating between expansion and contraction. NEG- t -SNE is consequently less sensitive to learning-rate annealing, and its clusters remain less fuzzy even when $\lambda = 1.0$ is held constant. Its repulsion shape is also bounded within $[0, 1]$ and does not diverge as $\zeta \rightarrow 0$, so fewer randomly sampled points are ejected from clusters. Whereas Damrich et al. (2023) explained this disparity through bounded NEG- t -SNE repulsion versus unbounded UMAP repulsion, our shape analysis in Fig. 1 shows that attraction is equally responsible for stability and compactness. Appendix F provides additional NEG- t -SNE discussion and illustrations. Together with Fig. 4 this result makes a concrete design principle: attraction and repulsion shapes can be mixed across methods, for example by pairing UMAP attraction with NEG- t -SNE repulsion. Appendix J demonstrates this principle on UMAP, NEG- t -SNE, and PaCMAP.

7 Discussion and Conclusion

This work establishes a force-shape account of how attractive and repulsive forces create clusters in neighbor embeddings. The standard description says that attraction brings similar points together and repulsion pushes dissimilar points apart, but that description does not explain the mechanisms that produce the final visualization. By analyzing the shapes of the forces, we make those mechanisms explicit. Although the main text focuses on UMAP and NEG- t -SNE, the framework applies broadly to dimensionality reduction, with additional algorithms analyzed in Appendix G.

We treat attraction and repulsion coefficients as functional mappings and analyze their shapes directly. The resulting characterization (Propositions 4.1 and 4.2) reveals a decisive and counterintuitive fact: UMAP’s attractive force expands short distances (Fig. 1(b)) instead of always pulling neighbors together. UMAP’s learning-rate schedule is therefore not merely an optimization convenience; it is the mechanism that suppresses this expansion.

We also identify why initialization matters. Similar points that start far apart experience weak attraction and may never contract sufficiently. The shape formalism provides a direct remedy: add terms that strengthen attraction at larger

distances. This modification produces more consistent outputs under random initialization, supporting the conclusion that UMAP encodes a reproducible structure and that standard initialization strategies help the optimization recover it efficiently.

The repulsion analysis shows that stronger repulsion increases inter-cluster distance. Adding a small positive value to the repulsion shape gives a level of repulsion that the traditional minimum-distance parameter m_d does not provide directly.

Finally, the comparison with NEG- t -SNE clarifies the role of bounded shapes. For $a = 1$ and $b = 1$, NEG- t -SNE differs from UMAP by providing bounded attraction ($\in [-1, 0]$) and repulsion ($\in [0, 1]$) shapes, so both propositions hold across learning rates in $[0, 1]$. This explains its reduced sensitivity to a fixed learning rate. Changing a and b in NEG- t -SNE reintroduces the same numerical challenges seen in UMAP. We also show that NEG- t -SNE and Parametric UMAP are equivalent.

These attraction-repulsion insights provide practical tools for designing and optimizing dimensionality reduction algorithms. Because dimensionality reduction is closely connected to contrastive learning (Damrich and Hamprecht, 2021; Hu et al., 2023), the same force-shape perspective can also inform representation learning. Overall, the framework turns embedding interpretation from a heuristic reading of pictures into a more principled analysis of the forces that generate them.

Software and Data

All data used in this paper are publicly available. The MNIST dataset is available at yann.lecun.com.

Fashion-MNIST is available at github.com.

Single-cell transcriptomes data is available at github.com.

References

- Akshay Agrawal, Alnur Ali, Stephen Boyd, et al. Minimum-distortion embedding. *Foundations and Trends® in Machine Learning*, 14(3):211–378, 2021.
- Mahsun Altin and Altan Cakir. Exploring the influence of dimensionality reduction on anomaly detection performance in multivariate time series. *IEEE Access*, 2024.
- Ehsan Amid and Manfred K Warmuth. TriMap: Large-scale dimensionality reduction using triplets. *arXiv preprint arXiv:1910.00204*, 2019.
- Sanjeev Arora, Wei Hu, and Pravesh K Kothari. An analysis of the t-sne algorithm for data visualization. In *Conference On Learning Theory*, pages 1455–1462. PMLR, 2018.
- Etienne Becht, Leland McInnes, John Healy, Charles-Antoine Dutertre, Immanuel WH Kwok, Lai Guan Ng, Florent Ginhoux, and Evan W Newell. Dimensionality reduction for visualizing single-cell data using umap. *Nature biotechnology*, 37(1):38–44, 2019.
- Mikhail Belkin and Partha Niyogi. Laplacian eigenmaps and spectral techniques for embedding and clustering. In *Advances in neural information processing systems*, pages 585–591, 2002.
- Jan Niklas Böhm, Philipp Berens, and Dmitry Kobak. Attraction-repulsion spectrum in neighbor embeddings. *The Journal of Machine Learning Research*, 23(1):4118–4149, 2022.
- Ingwer Borg and Patrick JF Groenen. *Modern multidimensional scaling: Theory and applications*. Springer Science & Business Media, 2007.
- T Tony Cai and Rong Ma. Theoretical foundations of t-sne for visualizing high-dimensional clustered data. *Journal of Machine Learning Research*, 23(301):1–54, 2022.
- Tara Chari and Lior Pachter. The specious art of single-cell genomics. *PLOS Computational Biology*, 19(8):e1011288, 2023.
- Sebastian Damrich and Fred A Hamprecht. On UMAP’s true loss function. *Advances in Neural Information Processing Systems*, 34:5798–5809, 2021.
- Sebastian Damrich, Niklas Böhm, Fred A Hamprecht, and Dmitry Kobak. From t -SNE to UMAP with contrastive learning. In *The Eleventh International Conference on Learning Representations*, 2023.

- Cyril de Bodt, Alex Diaz-Papkovich, Michael Bleher, Kerstin Bunte, Corinna Coupette, Sebastian Damrich, Enrique Fita Sanmartin, Fred A Hamprecht, Emőke-Ágnes Horvát, Dhruv Kohli, et al. Low-dimensional embeddings of high-dimensional data. *arXiv preprint arXiv:2508.15929*, 2025.
- Andrew Draganov and Simon Dohn. Relating tsne and umap to classical dimensionality reduction. *arXiv e-prints*, pages arXiv–2306, 2023.
- Andrew Draganov, Jakob Jørgensen, Katrine Scheel, Davide Mottin, Ira Assent, Tyrus Berry, and Cigdem Aslay. ActUp: Analyzing and consolidating tSNE and UMAP. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI-23*, pages 3651–3658, 8 2023.
- Shu-Kai S Fan, Du-Ming Tsai, Chih-Hung Jen, Chia-Yu Hsu, Fei He, and Li-Ting Juan. Data visualization of anomaly detection in semiconductor processing tools. *IEEE Transactions on Semiconductor Manufacturing*, 35(2):186–197, 2021.
- Jason Fleischer and Mohammad Tariqul Islam. Late breaking abstract-identifying and phenotyping COVID-19 patients using machine learning on chest x-rays. *European Respiratory Journal*, 2020.
- Jonathan Foote. Visualizing music and audio using self-similarity. In *Proceedings of the seventh ACM international conference on Multimedia (Part 1)*, pages 77–80, 1999.
- Rita González-Márquez, Luca Schmidt, Benjamin M Schmidt, Philipp Berens, and Dmitry Kobak. The landscape of biomedical research. *Patterns*, 2024.
- John C Gower. Generalized procrustes analysis. *Psychometrika*, 40:33–51, 1975.
- Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare (HEALTH)*, 3(1):1–23, 2021.
- Geoffrey Hinton and Sam T Roweis. Stochastic neighbor embedding. In *Advances in Neural Information Processing Systems*, volume 15, pages 833–840, 2002.
- Tianyang Hu, Zhili Liu, Fengwei Zhou, Wenjia Wang, and Weiran Huang. Your contrastive learning is secretly doing stochastic neighbor embedding. In *The Eleventh International Conference on Learning Representations*, 2023.
- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning*, pages 20617–20642, 2024.
- Mohammad Tariqul Islam and Jason W Fleischer. Manifold-aligned neighbor embedding. In *ICLR 2022 Workshop on Geometrical and Topological Representation Learning*, 2022.
- Mohammad Tariqul Islam and Jason W Fleischer. Outlier detection in large radiological datasets using umap. In *International Workshop on Topology-and Graph-Informed Imaging Informatics*, pages 111–121. Springer, 2024.
- Robert A Jacobs. Increased rates of convergence through learning rate adaptation. *Neural networks*, 1(4):295–307, 1988.
- Mathieu Jacomy, Tommaso Venturini, Sebastien Heymann, and Mathieu Bastian. Forceatlas2, a continuous graph layout algorithm for handy network visualization designed for the gephi software. *PloS one*, 9(6):e98679, 2014.
- Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.
- Dmitry Kobak and George C Linderman. Initialization is critical for preserving global data structure in both t-sne and umap. *Nature biotechnology*, 39(2):156–157, 2021.
- Dmitry Kobak, George Linderman, Stefan Steinerberger, Yuval Kluger, and Philipp Berens. Heavy-tailed kernels reveal a finer cluster structure in t-sne visualisations. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 124–139. Springer, 2019.
- Alexander Kolpakov and Aidan Rocke. The information geometry of umap. *Le Matematiche*, 79(1):151–164, 2024.
- Nikita Kotlov, Kirill Shaposhnikov, Cagdas Tazearslan, Madison Chasse, Artur Baisangurov, Svetlana Podsvirova, Dawn Fernandez, Mary Abdou, Leznath Kaneunyenye, Kelley Morgan, et al. Procrustes is a machine-learning approach that removes cross-platform batch effects from clinical rna sequencing data. *Communications Biology*, 7(1):392, 2024.
- Joseph B Kruskal. Nonmetric multidimensional scaling: a numerical method. *Psychometrika*, 29(2):115–129, 1964.
- Noël Kury, Dmitry Kobak, and Sebastian Damrich. DREAMS: Preserving both local and global structure in dimensionality reduction. *arXiv preprint arXiv:2508.13747*, 2025.

- Siu Kwan Lam, Antoine Pitrou, and Stanley Seibert. Numba: A llvm-based python jit compiler. In *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC*, pages 1–6, 2015.
- Pierre Lambert, Cyril De Bodt, Michel Verleysen, and John A Lee. Squadmds: a lean stochastic quartet mds improving global structure preservation in neighbor embedding like t-sne and umap. *Neurocomputing*, 503:17–27, 2022.
- George C Linderman and Stefan Steinerberger. Clustering with t-SNE, provably. *SIAM Journal on Mathematics of Data Science*, 1(2):313–332, 2019.
- George C Linderman and Stefan Steinerberger. Dimensionality reduction via dynamical systems: The case of t-sne. *SIAM Review*, 64(1):153–178, 2022.
- Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of machine learning research*, 9 (Nov):2579–2605, 2008.
- Evan Z Macosko, Anindita Basu, Rahul Satija, James Nemesh, Karthik Shekhar, Melissa Goldman, Itay Tirosh, Allison R Bialas, Nolan Kamitaki, Emily M Martersteck, et al. Highly parallel genome-wide expression profiling of individual cells using nanoliter droplets. *Cell*, 161(5):1202–1214, 2015.
- Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018.
- Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. Distributed representations of words and phrases and their compositionality. In *Advances in neural information processing systems*, pages 3111–3119, 2013.
- Ashwin Narayan, Bonnie Berger, and Hyunghoon Cho. Assessing single-cell transcriptomic variability through density-preserving data visualization. *Nature Biotechnology*, 39(6):765–774, 2021.
- F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- Pavlin G. Poličar, Martin Stražar, and Blaž Zupan. opentsne: A modular python library for t-sne dimensionality reduction and embedding. *Journal of Statistical Software*, 109(3):1–30, 2024.
- Aditya Ravuri and Neil D Lawrence. Towards one model for classical dimensionality reduction: A probabilistic perspective on umap and t-sne. *arXiv preprint arXiv:2405.17412*, 2024.
- Aditya Ravuri, Francisco Vargas, Vidhi Lalchand, and Neil D Lawrence. Dimensionality reduction as probabilistic inference. In *ICML 2023 Workshop on Structured Probabilistic Inference & Generative Modeling*, 2023.
- Peter J Rousseeuw. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of computational and applied mathematics*, 20:53–65, 1987.
- Sam T Roweis and Lawrence K Saul. Nonlinear dimensionality reduction by locally linear embedding. *science*, 290 (5500):2323–2326, 2000.
- Tim Sainburg, Leland McInnes, and Timothy Q Gentner. Parametric umap embeddings for representation and semisupervised learning. *Neural Computation*, 33(11):2881–2907, 2021.
- Uri Shaham and Stefan Steinerberger. Stochastic neighbor embedding separates well-separated clusters. *arXiv preprint arXiv:1702.02670*, 2017.
- Xintong Shi, Wenzhi Cao, and Sebastian Raschka. Deep neural networks for rank-consistent ordinal regression based on conditional probabilities. *Pattern Analysis and Applications*, 26(3):941–955, 2023.
- Jian Tang, Jingzhou Liu, Ming Zhang, and Qiaozhu Mei. Visualizing large-scale and high-dimensional data. In *Proceedings of the 25th international conference on world wide web*, pages 287–297, 2016.
- Laurens Van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of machine learning research*, 9 (11), 2008.
- Jarkko Venna and Samuel Kaski. Neighborhood preservation in nonlinear projection methods: An experimental study. In *International Conference on Artificial Neural Networks*, pages 485–491. Springer, 2001.
- Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.

- Yingfan Wang, Haiyang Huang, Cynthia Rudin, and Yaron Shaposhnik. Understanding how dimension reduction tools work: an empirical approach to deciphering t-SNE, UMAP, TriMAP, and PaCMAP for data visualization. *The Journal of Machine Learning Research*, 22(1):9129–9201, 2021.
- Yingfan Wang, Yiyang Sun, Haiyang Huang, and Cynthia Rudin. Dimension reduction with locally adjusted graphs. In *Thirty-Ninth AAAI Conference on Artificial Intelligence*, 2025.
- Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. 2017.
- Yang Yang, Hongjian Sun, Jialei Gong, Yali Du, and Di Yu. Interpretable dimensionality reduction by feature preserving manifold approximation and projection. *arXiv preprint arXiv:2211.09321*, 2022.
- Lihi Zelnik-Manor and Pietro Perona. Self-tuning spectral clustering. *Advances in neural information processing systems*, 17, 2004.
- Jonathan X Zheng, Samraat Pawar, and Dan FM Goodman. Graph drawing by stochastic gradient descent. *IEEE transactions on visualization and computer graphics*, 25(9):2738–2748, 2018.

Appendix

Appendix [A](#) provides the derivations for UMAP attraction and repulsion shapes. Appendix [B](#) extends the main-text analysis of constant learning rates. Appendix [C](#) demonstrates how stronger attraction and repulsion affect large language model embeddings. Appendix [D.1](#) formally defines the Procrustes distance and matrix used to quantify consistency. Appendix [E](#) confirms the random-initialization results on additional datasets. Appendix [F](#) expands the NEG- t -SNE and UMAP comparison. Appendix [G](#) applies the same force-shape view to other dimensionality reduction methods, and Appendix [H](#) records the high-dimensional graph construction. Appendix [I](#) gives the detailed sweeps over λ_a and λ_r (from Fig. [11](#)) for UMAP, NEG- t -SNE, and PaCMAP across datasets. Appendix [J](#) shows that attraction and repulsion shapes can be combined across algorithms. Appendix [K](#) reports the implementation details.

A Proofs and Derivations

A.1 Proof of Proposition [4.1](#)

Proof. Subtracting Eq. [\(7\)](#) from Eq. [\(8\)](#) and taking a norm gives

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

The distance contracts exactly when $|1 + 2\lambda f_a| < 1$, i.e., when

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition [4.2](#)

Proof. Subtracting Eq. [\(9\)](#) from Eq. [\(10\)](#) and taking a norm gives

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

The distance increases when $|1 + \lambda f_r| > 1$, i.e.,

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

By the definition of f_r , the latter condition suffices. □

A.3 Attraction Shape:

We derive the attraction shape from the general low-dimensional affinity $q_{ij} = (\gamma + a\|y_i - y_j\|_2^{2b})^{-1}$, which reduces to UMAP when $\gamma = 1$ and to NEG- t -SNE when $\gamma = 2$. The attractive force is

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

Defining $\zeta = \|y_i - y_j\|_2$ isolates the attraction shape as

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

A.4 Condition for strictly increasing f_a^U :

The distance dependence of f_a^U is determined by its derivative:

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

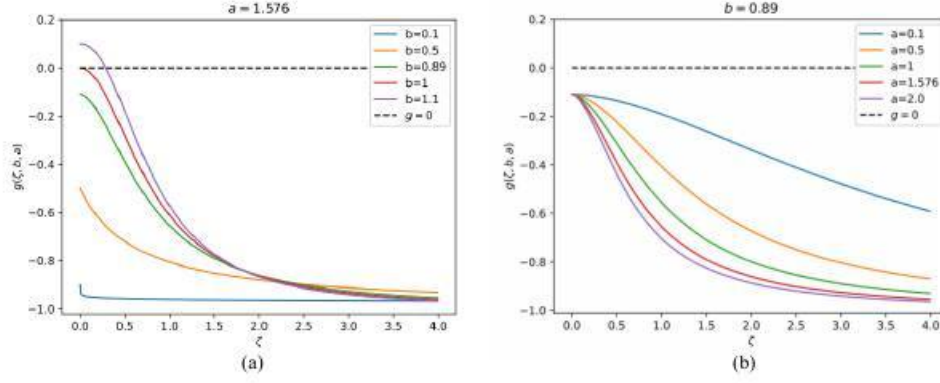


Figure 5: Values of $g(\zeta, b, a)$ demonstrate the monotonicity condition for (a) fixed $a = 1.576$ and (b) fixed $b = 0.89$.

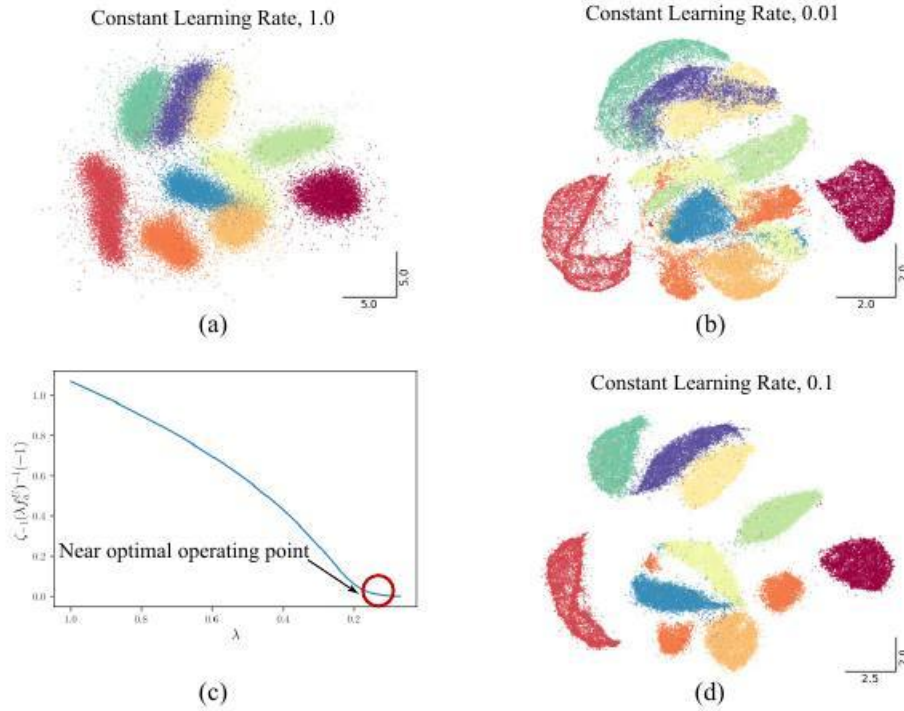


Figure 6: Constant learning rates expose the tradeoff controlled by ζ_{-1} . (a) A high learning rate ($\lambda = 1.0$) yields diffuse embeddings because ζ_{-1} is large. (b) A very low learning rate ($\lambda = 0.01$) prevents cluster formation because attraction and repulsion are too weak. (c) ζ_{-1} decreases nonlinearly with the learning rate. The effective objective is to reduce ζ_{-1} while preserving sufficient attraction and repulsion. (d) A near-optimal constant learning rate $\lambda = 0.1$ produces distinct, compact clusters.

This yields the strictly increasing condition ($\frac{df_a^U}{d\zeta} > 0$),

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. This inequality holds for $0 < b \leq 1$ by the derivative and asymptotes of g . Figure 5 shows values of g for different b and a , and it confirms that the inequality (23) fails once $b > 1$.

A.5 Repulsion Shape:

Using the same general low-dimensional affinity, the repulsive force is

$$\begin{aligned}
 \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\
 &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j).
 \end{aligned} \tag{24}$$

The scalar term gives the repulsion shape as

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}. \tag{25}$$

Thus, $f_r > 0$ in the relevant setting.

A.6 Loss functions due to modified attraction and repulsion shape:

The cost function for the attractive term modified in Eq. (13) is

$$- \int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j) dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \tag{26}$$

and the repulsive term modified in Eq. (15) is

$$- \int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j) dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \tag{27}$$

In both cases, the added term is a norm-based regularizer. Expressing the modification at the shape level makes the effect explicit: the attractive term increases far-sightedness, while the repulsive term adds a constant repulsive coefficient.

B More on UMAP’s Learning Rate

Sections 5 and 6 show why UMAP depends on learning-rate annealing (Fig 6). Section 5 defines the minimum contraction distance ζ_{-1} and establishes that annealing reduces this value, thereby enabling compact clusters. Section 6 compares the attraction shapes of UMAP (for $a = 1.0$ and $b = 1.0$) and Neg- t -SNE and shows why Neg- t -SNE tolerates a constant learning rate of 1.0 better: its attraction shape remains within $[-1, 0]$, whereas UMAP’s lies within $[-2, 0]$. By the same logic, UMAP tolerates a constant learning rate of 0.5 when the effective attraction shape is confined to $[-1, 0]$ (Figs. 11 f,h).

For a wide range of a and b , annealing still improves the final embedding. The reason is mechanistic: the algorithm must eventually push ζ_{-1} toward zero, which happens as the learning rate approaches zero. If the learning rate remains too high, the embedding is diffuse (Fig 6a). If it starts too low, attraction and repulsion are too weak to form clear clusters (Fig 6b). The ζ_{-1} versus λ curve (Fig 6c) identifies $\lambda = 0.1$ as a near-optimal constant value, balancing a low ζ_{-1} with substantial attraction and repulsion. At this constant learning rate, the embedding forms compact clusters with clear boundaries (Fig 6d).

C Large Language Model Embedding

Stronger attraction and repulsion can improve embeddings in large language model feature spaces. To demonstrate this, we randomly selected 300,000 PubMedBERT (Gu et al. 2021) feature vectors with known labels from the PubMed dataset (González-Márquez et al. 2024). Default UMAP produces a crowded structure with overlapping label colors (Fig. 7a). Without labels, the structure is difficult to interpret because explicit clusters are weak. Setting $b = 0.6$ as motivated by Section 5.2 increases repulsion while also strengthening attraction-driven cluster formation. The resulting mapping is interpretable even without explicit labels (Fig. 7b). The k-NN accuracy ($k = 10$) quantifies the improvement. On a 10,000-point hold-out set, reducing b to 0.6 increases k-NN accuracy from 49.2% to 55.48%.

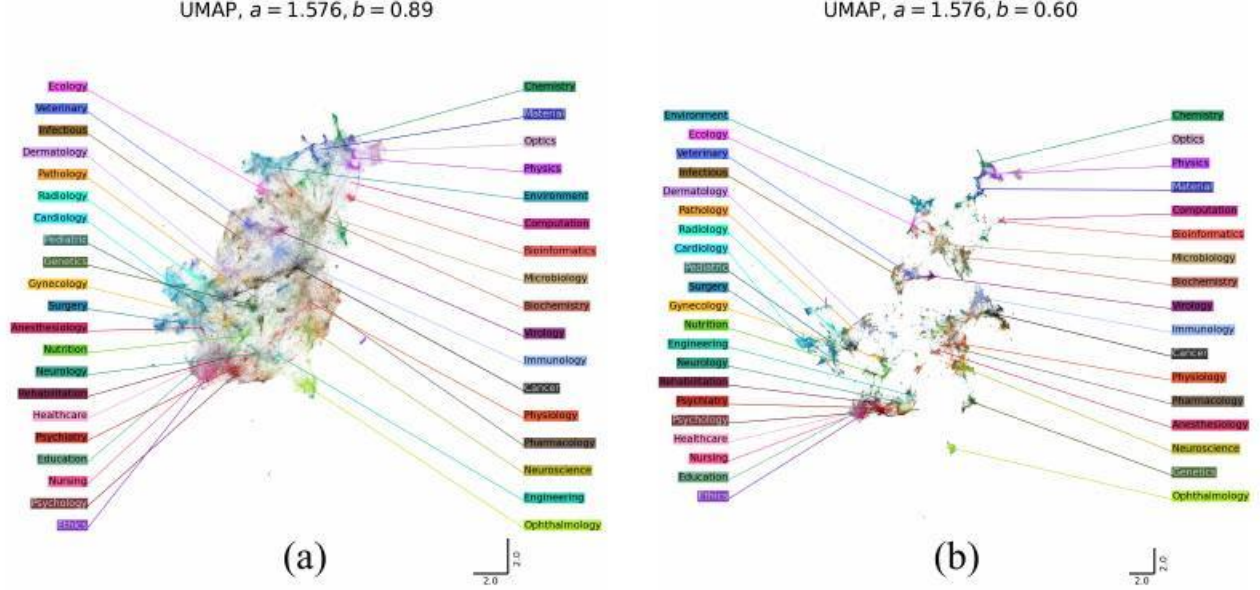


Figure 7: Visualization of 300K PubMed abstracts. (a) Default UMAP ($a = 1.576, b = 0.89$) produces overlapping labels in the mapping. (b) UMAP with stronger attraction and repulsion ($a = 1.576, b = 0.60$) yields clearer cluster formation and non-overlapping labels. Labels are shown in both mappings for interpretability.

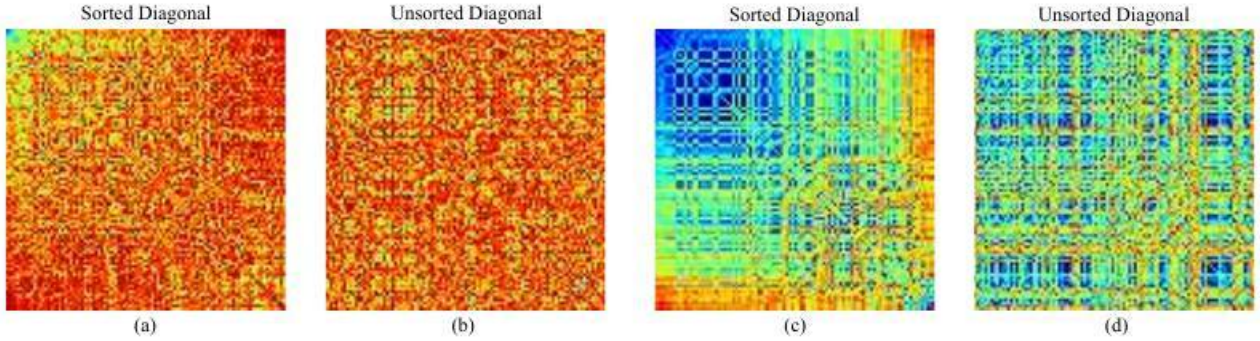


Figure 8: Sorting the Procrustes matrix diagonal reveals self-similarity. (a) Procrustes matrix from [2]f) for default UMAP attraction, with the diagonal sorted by p_d from the PCA-initialized reference. (b) The same data with an unsorted (randomly ordered) diagonal. (c) Procrustes matrix from [2]g) for modified attraction, again sorted by p_d from the PCA-initialized reference. (d) The same data as in (c) with an unsorted diagonal.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

The Procrustes distance (Gower, 1975) measures similarity between two point clouds $\{x\}$ and $\{y\}$ after translation, scaling, and rotation. Operationally, we hold $\{x\}$ fixed and transform $\{y\}$ to maximize alignment. Let $\{y'\}$ denote the aligned version of $\{y\}$. The Procrustes distance is

$$p_d(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

The Procrustes distance is a linear measure that is useful across embedding and alignment settings (McInnes et al., 2018; Islam and Fleischer, 2022; Kotlov et al., 2024).

We use Procrustes distance to measure embedding consistency under random initialization. Let $\{x\}_p$ be a reference embedding (PCA initialization in our experiments), and let $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ be N embeddings obtained

from random initialization. Embedding similarity is summarized by the mean and standard deviation of the strictly lower triangular values of the matrix P (reported as mean $\pm$ std in Figs. 2, 9, and 10), with

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

We sort the indices of X_r so that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$, making the diagonal values of the Procrustes matrix

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal values

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

The diagonal sorting adds little numerically, but it reveals self-similarity visually. In Fig. 8(a), the sorted diagonal exposes a cluster of similar embeddings in the upper-left region of the matrix. With the modified attraction shape, the number of mutually similar embeddings increases, as shown by the larger blue region in Fig. 8(c), indicating a metastable point in the embedding algorithm. When the diagonal is unsorted, this structure disappears and the visual evidence of similar embeddings is lost (Fig. 8(b,d)).

D.2 Trustworthiness

The trustworthiness metric (Venna and Kaski, 2001) quantifies local-neighborhood preservation after dimensionality reduction:

$$T = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ is the k -NN graph in the embedding space and $r(i, j)$ is the rank of x_j in the high-dimensional k -NN graph. We set $k = 5$ in practice to assess preservation of each point's five nearest neighbors. For computational efficiency, reported trustworthiness values use 10,000 randomly sampled indices, with the same indices used across embeddings being compared.

D.3 Silhouette Score

Although the silhouette score (Rousseeuw, 1987) is designed for clustering algorithms, we use it to evaluate label separation in embeddings, namely how well ground-truth labels separate into clusters. This is appropriate here because the embedding algorithms explicitly create clustered geometry and should separate labels when the structure supports it. For a point y_i in a point cloud $\{y\}$, let a_i be the mean distance from y_i to points with the same label, and let b_i be the minimal mean distance from y_i to points in any other label. The pointwise silhouette is

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This value lies within $[-1, 1]$. Values near 1 indicate that y_i fits well within its label cluster, values near 0 indicate a boundary point, and values near -1 indicate failed label separation. The overall silhouette score is

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

We compute the silhouette score on the full embedding, without random sampling, using Euclidean distances.

E Effect of Random Initialization in Additional Datasets

The main text (Section 5.1) reports the random-initialization experiment on MNIST. Here we repeat the same experiment on Fashion-MNIST (FMNIST) (Xiao et al., 2017) and single-cell transcriptomes (Macosko et al., 2015) data (Fig. 9 and 10 respectively). The conclusion is robust: modified and composite attraction shapes that increase long-distance attraction substantially improve reconstruction consistency.

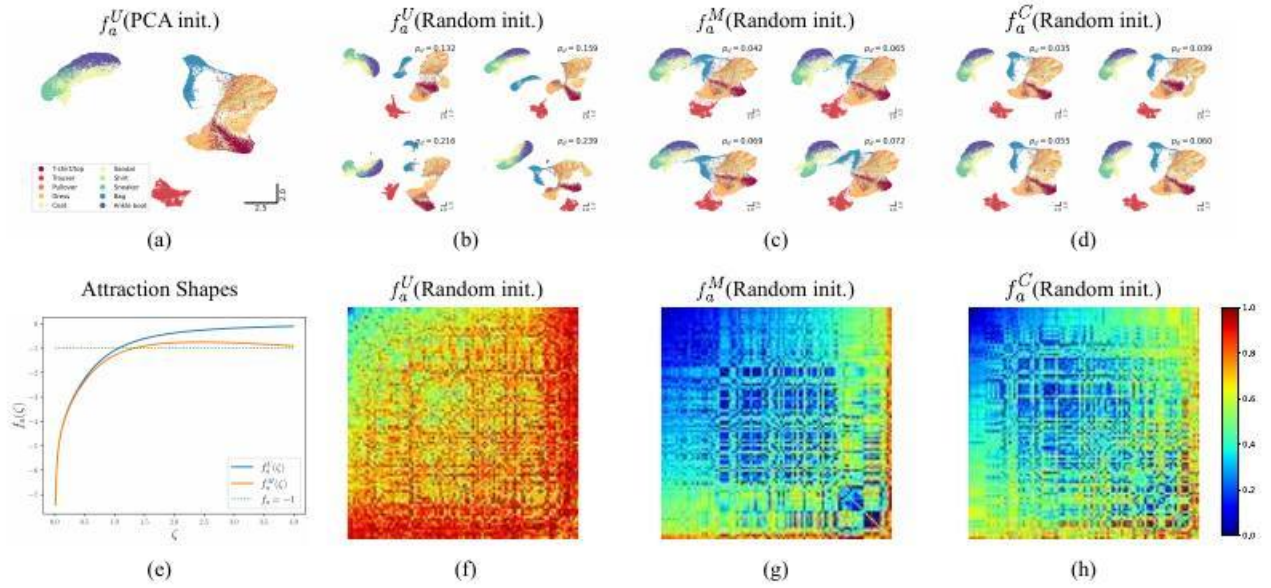


Figure 9: Random UMAP initialization under different attraction shapes on FMNIST. (a) PCA-initialized reference mapping. (b-d) Four mappings with the lowest Procrustes distance (p_d) from (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs for (f) default (0.72 ± 0.15), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18) attraction shapes.

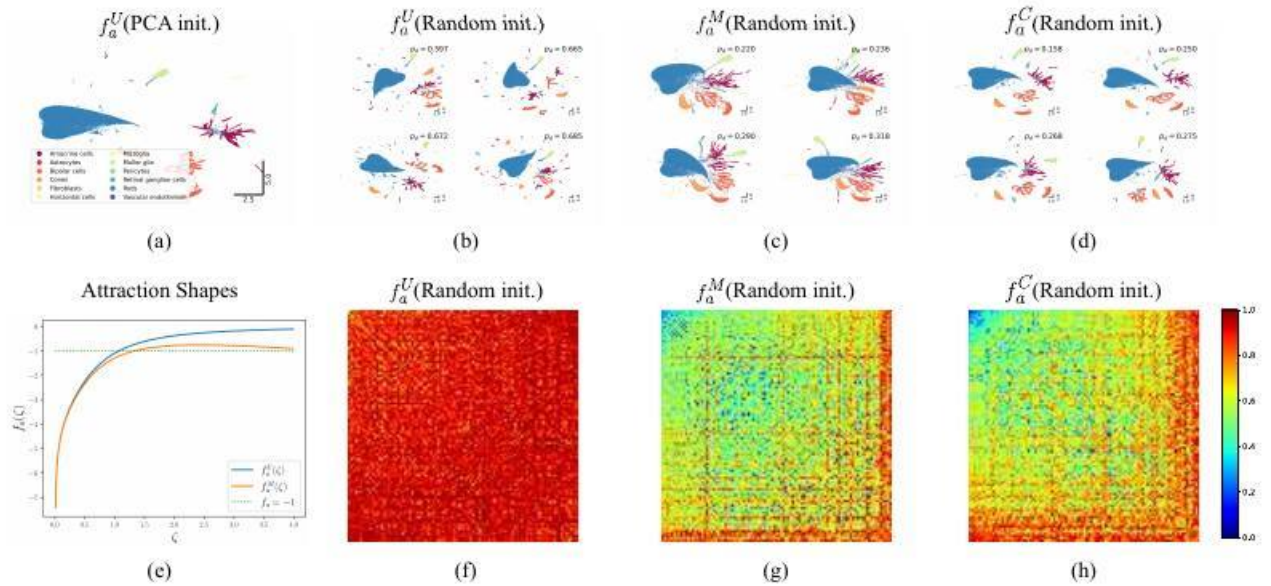


Figure 10: Random UMAP initialization under different attraction shapes on single-cell transcriptomes data. (a) PCA-initialized reference mapping. (b-d) Four mappings with the lowest Procrustes distance (p_d) from (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs for (f) default (0.91 ± 0.06), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15) attraction shapes.

F More on NEG- t -SNE

F.1 Comparison to UMAP

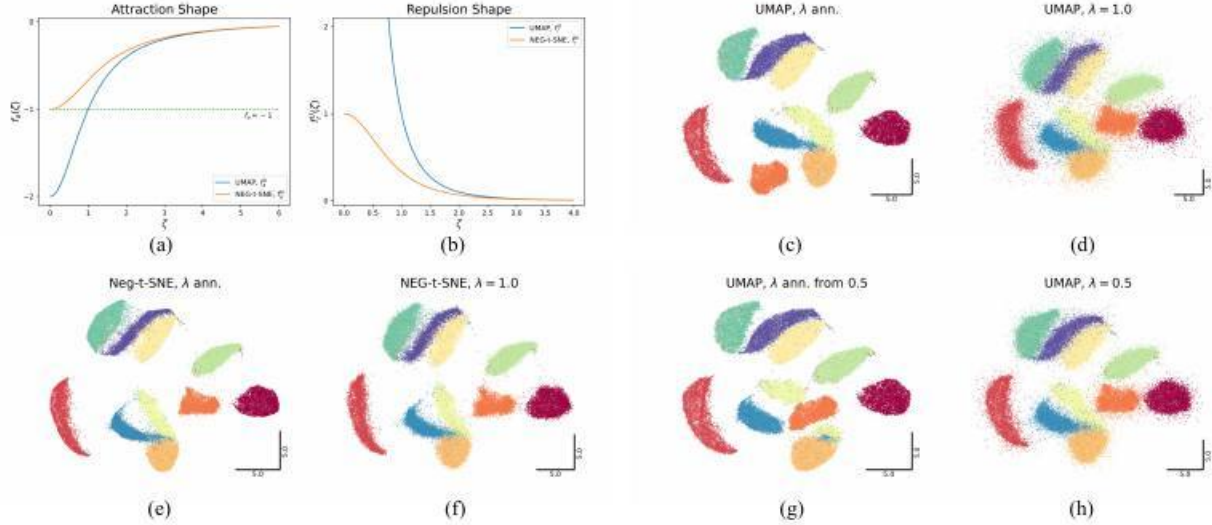


Figure 11: UMAP and NEG- t -SNE differ sharply in learning-rate sensitivity on MNIST. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1$, $b = 1$) and NEG- t -SNE. (c,d) UMAP is highly sensitive to λ because $f_a^U < -1$ as separation distance ζ decreases; without annealing, clusters become fuzzy. (e,f) NEG- t -SNE is less sensitive because $f_a^N \in [-1, 0]$ always, so clusters remain less fuzzy without annealing. (g,h) Confining f_a^U to $[-1, 0]$ by setting $\lambda = 0.5$ reduces UMAP’s sensitivity to λ .

Figure 11 compares the shapes of UMAP and NEG- t -SNE together with their MNIST embeddings. With learning-rate annealing, UMAP and NEG- t -SNE produce similar outputs (Figs. 11 c,e). At a constant learning rate of 1, however, UMAP becomes fuzzy whereas NEG- t -SNE retains much cleaner boundaries (Fig. 11 d,f). Section 6 explains this behavior: constraining f_a^U to $[-1, 0]$ should reduce fuzziness at fixed λ . Fig. 11 h,i already supports this prediction, since UMAP gives better embedding and clustering when the attraction learning rate (λ_a) is $\lesssim 0.5$. The simplest implementation is to initialize λ at 0.5, which satisfies Proposition 4.1 for all ζ . The embeddings in Figs. 11 (g) and (h) confirm the prediction: with constant $\lambda = 0.5$, UMAP clusters become sharper and resemble those of NEG- t -SNE more closely (Fig. 11 d,h). A few points still remain outside clusters because of UMAP’s repulsion shape, a problem addressed by NEG- t -SNE.

We can also introduce a and b into NEG- t -SNE, which is essentially the Parametric UMAP formulation in Proposition 6.1. The affinity becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, and the attraction and repulsion shapes become $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2+a\zeta^{2b}}$, and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1+a\zeta^{2b})(2+a\zeta^{2b})}$, respectively. For $0 < b < 1$, both shapes become unbounded as $\zeta \rightarrow 0$. Thus, varying a and b makes NEG- t -SNE inherit UMAP-like numerical challenges and their associated limitations. The main distinction is that NEG- t -SNE reaches a lower minimum contraction distance (ζ_{-1}) for attraction than UMAP. This can enhance cluster formation, but the attraction also decays faster toward zero at long distances, increasing near-sightedness and potentially weakening long-range attraction.

F.2 Comparison to Parametric UMAP

Parametric UMAP was initially trained with the original UMAP objective (Sainburg et al. 2021), but later work adopted a numerically stable, log-sigmoid-based modified cross-entropy loss (Shi et al. 2023). This modification makes Parametric UMAP and Neg- t -SNE (Damrich et al. 2023) equivalent (Proposition 6.1). The proof below makes that equivalence explicit.

Proof of Proposition 6.1

Proof. The kernel function under this modification becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

And the repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Both these are NEG-t-SNE with explicit parameters a and b (while in Neg-t-SNE these are set to 1). $\square$

G Alternate Dimensionality Reduction Algorithms

The alternative algorithms considered here use the same low-dimensional kernel as UMAP when $a = 1$ and $b = 1$:

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

We first analyze TriMap (Amid and Warmuth 2019). Although TriMap uses triplets rather than pairwise interactions, it provides a useful bridge to attraction-repulsion mechanisms that are more coupled than in UMAP. We then analyze Pairwise Controlled Manifold Approximation (PaCMAP) (Wang et al. 2021) and its extension, Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP) (Wang et al. 2025), which adapt the TriMap logic to pairwise interactions. We next treat t -SNE (Van der Maaten and Hinton 2008), briefly discuss SNE (Hinton and Roweis 2002) with its alternate kernel, and conclude with multidimensional scaling (Borg and Groenen 2007).

G.1 TriMap

TriMap (Amid and Warmuth 2019) optimizes the low-dimensional embedding with a triplet loss

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} is the weight of the triplet (y_i, y_j, y_k) , y_j belongs to the high-dimensional k -nearest-neighbor set of y_i , and y_k is a far-away point. Minimization should attract y_i and y_j while repelling y_i and y_k . The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

where $\zeta_1 = \|y_i - y_j\|_2$ is the nearest-neighbor distance, $\zeta_2 = \|y_i - y_k\|_2$ is the distance to the far-away point, and f_a^T and f_r^T are TriMap's attraction and repulsion shapes. Unlike UMAP, the attractive and repulsive components are not separable, and each shape depends on two distances. The attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

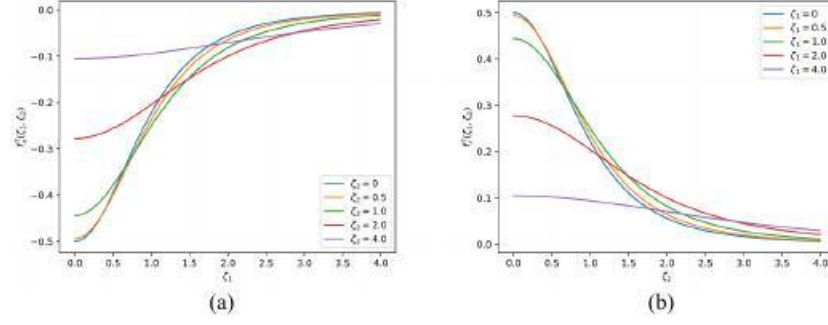


Figure 12: TriMap's coupled shapes. (a) Attraction shapes for different ζ_2 . (b) Repulsion shapes for different ζ_1 .

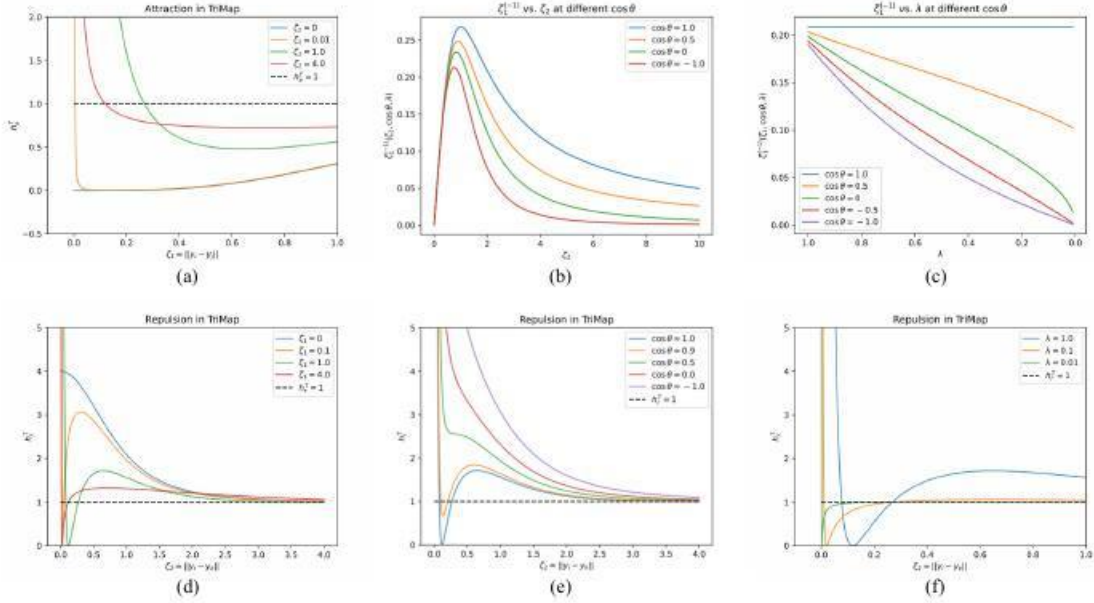


Figure 13: TriMap's attraction and repulsion behavior. (a) h_a^T versus ζ_1 for different ζ_2 ; values below the dotted line indicate attraction. Like UMAP, TriMap can show attraction and repulsion for nearest neighbors (y_i, y_j). (b,c) Unlike UMAP, the minimum contraction distance ($\zeta_1^{(-1)}$) varies with (b) ζ_2 and (c) λ ; $\cos \theta$ controls the range of these values. (d-f) Repulsion in TriMap when varying (d) ζ_1 , (e) $\cos \theta$, and (f) λ . Values above (below) the dotted line indicate repulsion (attraction). Unlike UMAP's repulsive force, TriMap's repulsive component can produce both attraction and repulsion. Unless otherwise labeled, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

TriMap's attraction and repulsion shapes (Fig. 12) follow trends similar to UMAP's. The minimum attraction value is -0.5 , so, unlike UMAP, TriMap does not flip positions through attraction alone. However, Eqs. (45)-(47) couple attraction and repulsion, so Propositions 4.1 and 4.2 do not apply directly. Focusing first on attraction, we obtain

Proposition G.1. *Update equations (45)-(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i^t - y_j^t)^T (y_i^t - y_k^t)}{\|y_i^t - y_j^t\|_2 \|y_i^t - y_k^t\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46): $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T(y_i^t - y_k^t)$. Putting this value in Eq. (51), we obtain the desired inequality. $\square$

Inequality (50) depends jointly on ζ_1 , ζ_2 , $\cos \theta$, and λ . Thus, the target contraction distance $\zeta_1 = \|y_i - y_j\|_2$ is coupled to additional geometric variables. Figure 13a) shows the attraction behavior for different ζ_2 values with $\cos \theta = 1$ and $\lambda = 1$. Values below the dotted line indicate attraction and contraction of ζ_1 , whereas values above indicate repulsion and expansion of ζ_1 . The intersection of the dotted line with h_a^T gives the minimum contraction distance ($\zeta_1^{(-1)}$), analogous to UMAP's ζ_{-1} , which we define as

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

$\zeta_1^{(-1)}$ is finite and positive in most cases (Fig. 13b,c)). TriMap can therefore exhibit UMAP-like behavior and require learning-rate annealing or an analogous adaptive strategy; the original implementation uses delta-bar-delta (Jacobs, 1988) with appropriate initialization.

Proposition G.2. *Update equations (45)-(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T(1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47): $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T(y_i^t - y_j^t)$. Putting this value in Eq. (55) we obtain inequality (54). $\square$

Inequality (54) also depends on ζ_1 , ζ_2 , $\cos \theta$, and λ . Here the quantity of interest is the expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13d-f) show the repulsion behavior under variations of the other quantities. Values above the dotted line indicate repulsion and expansion, while values below indicate attraction. The key contrast with UMAP is decisive: TriMap's repulsive component can contract rather than expand. Because this anomaly appears at small distances, appropriate initialization and triplet selection can avoid it.

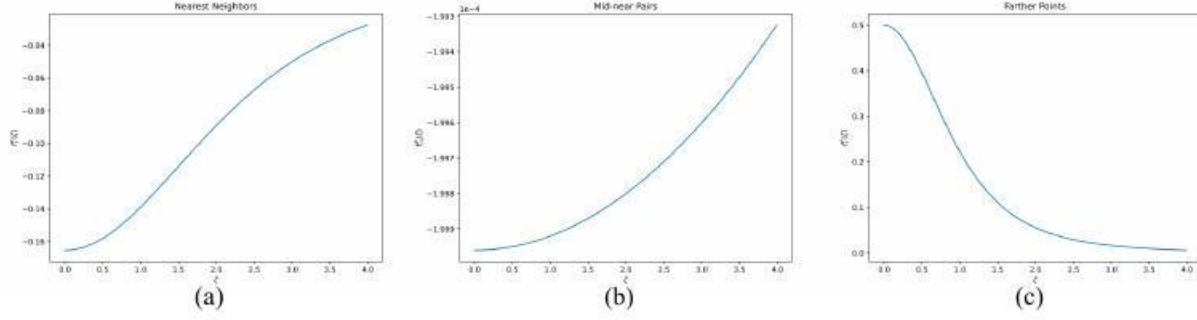


Figure 14: PaCMAP separates force roles through bounded shapes. (a,b) Attraction shapes for (a) nearest-neighbor and (b) mid-near points (values are on the order of 10^{-4}). (c) Repulsion shape for farther pairs. $\lambda = 1$ for all figures.

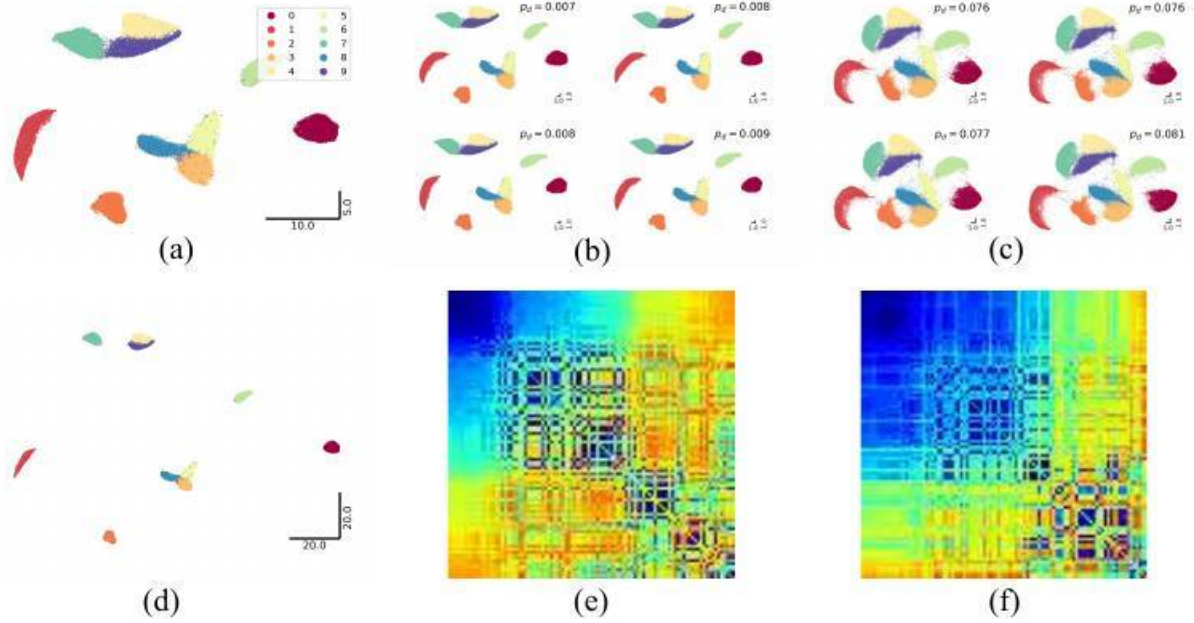


Figure 15: PaCMAP behavior under controlled shape modifications. (a) PCA-initialized PaCMAP embedding of MNIST. (b,c) Four random-initialized samples that best match (a) for (b) original PaCMAP and (c) PaCMAP with modified attraction $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$. (d) PaCMAP on MNIST with modified repulsion $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrices for (e) original PaCMAP (0.51 ± 0.22) and (f) modified-attraction PaCMAP (0.43 ± 0.22). As in UMAP, stronger long-distance attraction improves consistency, while stronger repulsion yields smaller clusters and larger inter-cluster distances.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al., 2021) optimizes the low-dimensional embedding across multiple scales. Its loss is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} weight nearest-neighbor (NN) pairs, mid-near (MN) pairs, and further pairs (FP), respectively (details in Appendix H). The first two terms provide attraction, and the final term provides repulsion. The loss is a modified pairwise version of TriMap’s triplet loss: the attractive terms replace TriMap’s affinity for distant points with constants ($1/10$ for nearest neighbors, $1/10000$ for mid-near pairs, and 1 for farther points). This structure

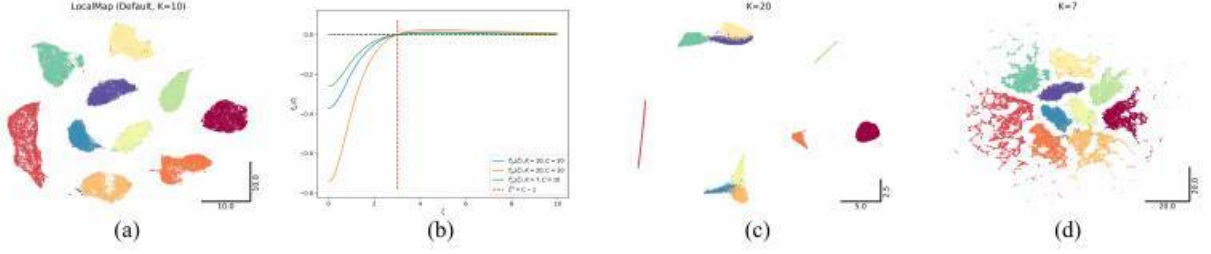


Figure 16: LocalMAP behavior on MNIST. (a) Default embedding. (b) Nearest-neighbor attraction-repulsion shape; values below (above) the dotted line indicate attraction (repulsion). The transition occurs when ζ crosses $\sqrt{C-1}$, with $C = 10$ as in the LocalMAP implementation. (c) Large K ($= 20$) makes repulsion dominate and compact clusters persist. (d) Small K ($= 7$) makes attraction dominate and breaks clusters apart.

separates the loss into three terms and decouples the update equations. The nearest-neighbor updates are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

the mid-near pair updates are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and the farthest-pair updates are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, and $\zeta_3 = \|y_i - y_l\|_2$ are distances; f_a^P and f_m^P are attraction shapes for nearest-neighbor and mid-near pairs; and f_r^P is the repulsion shape for farthest pairs. The shapes are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

f_a^P and f_m^P satisfy Proposition 4.1 and f_r satisfies Proposition 4.2 (Fig. 14). Attraction is weaker than in UMAP, but it remains effective over a wide learning-rate range when modulated by Adam (Kingma and Ba 2015); with $w_{NB} = 3$, the maximum attraction stays below 0.5, preventing flips during attractive updates. PaCMAP typically initializes the embedding inside a small low-dimensional sphere (e.g., on the order of 10^{-3} in (Wang et al. 2021)) and relies on repulsion to separate clusters. It largely recovers UMAP-like clustering properties, especially on MNIST, while improving ordering. Its random-initialization consistency already exceeds UMAP’s (Fig. 15 b,e) and improves further with modified attraction (Fig. 15 c,f). Adding a small value to the repulsion shape increases embedding compactness by increasing inter-cluster distance.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al. 2025) extends PaCMAP. A defining outcome is its separation of all 10 MNIST clusters (Fig. 16 a), with behavior paralleling Figs. 3 g,h,j,k) and Figs. 4 a,b). We use LocalMAP to further isolate how attraction and repulsion jointly control cluster compactness and connectedness.

LocalMAP first runs PaCMAP and then performs an additional optimization that changes nearest-neighbor interactions to decouple clusters. It minimizes

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term combines attractive and repulsive behavior on the same pair. In one regime it attracts, and in another it repels. The second term is the PaCMAP far-pair term, except that LocalMAP resamples further pairs every few iterations. Thus, the nearest-neighbor term is the decisive new component, with update equations

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_1^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_1^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the attraction-repulsion shape

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)^2}. \quad (69)$$

The update contracts when $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$ (from Proposition 4.1). When $\zeta^2 > C-1$, $f_{ar} > 0$, and Proposition 4.2 gives expansion. Thus, K and C determine whether attraction or repulsion dominates. The LocalMAP implementation sets both C and K to 10 (Fig. 16(b)), and its attraction is stronger than PaCMAP's (Fig. 14(a)). Consequently, when $\zeta > 3$, nearest neighbors experience repulsion, separating pairs that bridge two clusters. K acts as the force-scaling parameter that regulates this separation.

Using Proposition 4.1 $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ gives the maximum K and maximum attraction that f_{ar}^L can use without flipping pair placements. Avoiding flips is important because LocalMAP should preserve PaCMAP's ordering while separating clusters. This simplifies (69) to $K \leq \frac{\lambda(1+C)^2}{C-1}$, which gives $K \lesssim 13.44$ for $C = 10$ and $\lambda = 1$. For $K \gtrsim 13.44$, repulsion dominates and clusters become more compact, but the LocalMAP objective fails because the bridge between clusters persists (Fig. 16(c), $K = 20$). As K decreases, attraction dominates because repulsion is too weak, and existing clusters break apart (Fig. 16(d), $K = 7$), mirroring Fig. 4(b).

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton 2008) optimizes pairwise relationships through the loss

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the pair weight. The original t -SNE implementation considers all pairs, not only nearest neighbors. The loss decomposes into attraction and repulsion as

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As before, the first term provides attraction and the second provides repulsion. The attractive term matches UMAP with $a = 1$ and $b = 1$ and is straightforward to compute, whereas the repulsive term couples all pairs and is therefore expensive. Applying the same shape principle used for UMAP gives the t -SNE update equations. Because the original t -SNE does not rely on a nearest-neighbor graph, the all-pair weight $w_{i,j}$ is central in the updates. The attractive updates are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the t -SNE attraction shape and $\zeta_{i,j} = \|y_i - y_j\|_2$. The repulsive updates are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k}^t)(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j}^t)(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the t -SNE repulsion shape and $Z = \sum_{k \neq l} q_{k,l}$. The shapes are

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1+\zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1+\zeta^2)^2}. \quad (77)$$

From the attractive updates (72)-(73), f_a^{t-SNE} satisfies Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$) and defines a minimum contraction distance ζ_{-1} . The repulsive update is coupled across all pairs, so it does not reduce to a simple one-shape condition. Instead, using the same reasoning as in the TriMap analysis, we obtain the following:

Proposition G.3. *The update Eqs. (74)-(75) provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k,k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l,l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Substituting this value into Eq. (79) gives the desired inequality. $\square$

Because t -SNE’s repulsion condition (Eq. 78) mirrors TriMap’s, the same qualitative behavior follows: t -SNE repulsive forces can generate both expansion and contraction.

Because t -SNE scales its forces by $w_{i,j}$ for attraction and $\frac{w_{i,j}}{Z}$ for repulsion, its attractive and repulsive forces are typically weaker than UMAP’s. The algorithm therefore uses large learning rates, for example $\approx 10^3$ in Open- t -SNE (Poličar et al. 2024). Most t -SNE implementations also rely on an “early exaggeration” phase that multiplies attractive forces by a constant during the first iterations. This phase pulls far-apart points that should be neighbors toward each other, effectively inducing far-sightedness. It can also separate points that are too close ($\zeta_{i,j} \approx \zeta_{-1}$) but should be apart. Thus, “early exaggeration” is a central mechanism for obtaining consistent t -SNE embeddings, especially under random initialization. When applied throughout optimization, it makes t -SNE embeddings resemble UMAP embeddings more closely (Böhm et al. 2022).

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis 2002) is one of the earliest algorithms in this class. It uses the same loss form as t -SNE (Eq. 71) but sets $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$. This yields the attraction shape

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

which follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$), and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

which follows Proposition G.3 after replacing the t -SNE symbols with SNE counterparts. The attraction shape is poorly conditioned because contraction depends heavily on the learning rate (λ) and weights ($w_{i,j}$). This leads to overlapping clusters even with relatively few samples, the crowding problem (Van der Maaten and Hinton 2008). t -SNE and later algorithms address this issue by replacing the Gaussian kernel with a heavy-tailed kernel, as in Eq. (43).

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen 2007) usually avoids gradient methods because they often fail to converge to good mappings; stress majorization is the standard alternative. Some work nevertheless studies gradient formulations (Kruskal 1964; Zheng et al. 2018). We briefly analyze this case, focusing on the gradient-descent MDS algorithm for graph drawing in Zheng et al. (2018). We start from the cost function, ignoring $w_{i,j}$ without loss of generality for that approach and absorbing it into the learning rate for discussion:

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

The loss contains one term per pair and does not explicitly separate attraction from repulsion. Instead, each term itself can attract or repel. Because the loss separates into individual (i, j) components, we can analyze one pair directly. The derivative is

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

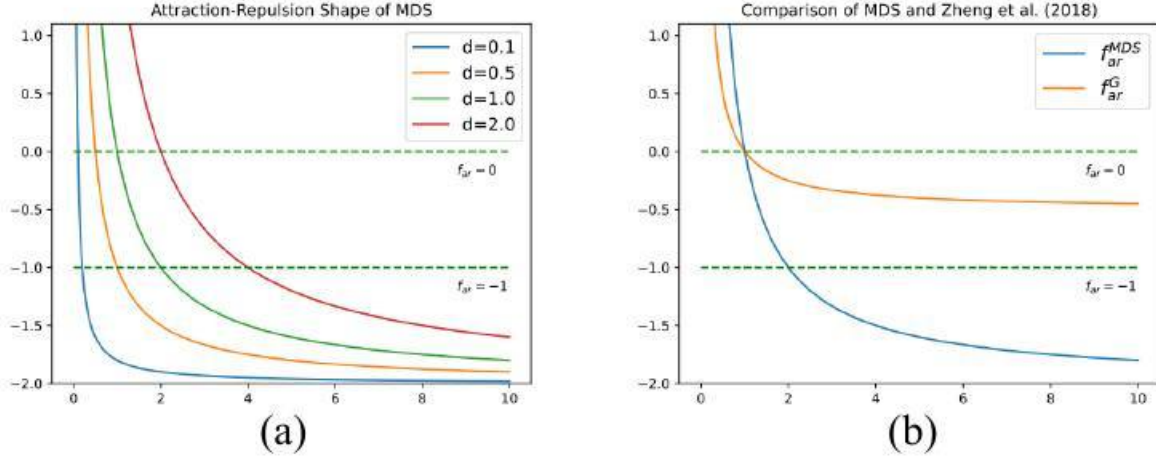


Figure 17: MDS attraction-repulsion behavior. (a) Attraction-repulsion characteristic of MDS using f_{ar} . (b) Comparison between MDS and the method in Zheng et al. (2018) for $d = 1.0$.

Thus, the MDS attraction-repulsion shape is

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. The shape is attractive when $\zeta > d$ and is bounded within $[-2, 0]$ in that range (Fig. 17(a)). This range is not uniformly suitable for convergence: only values in $[-1, 0]$ contract, while other values expand, and values below -2 flip the points and expand them. To avoid this, Zheng et al. (2018) uses a force-directed-graph-inspired formulation,

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

which works for an effective learning rate $\lambda \leq 1$. Since f_{ar}^G is bounded within $[-0.5, 0]$, $\lambda \leq 1$ gives stable contraction. For $\zeta < d$, $f_{ar}^{MDS}(> 0)$ gives repulsion. This attraction-repulsion interaction works best when initialization is already close to the desired output. Continued optimization of a pair oscillates around $\zeta = d$, where $f_{ar}^G = 0$, giving a distance threshold analogous to UMAP's ζ_{-1} . No learning rate drives this distance to zero, because preserving it is the MDS objective. Consequently, MDS clusters are often fuzzier than those from UMAP or PaCMAP (for illustrations, see Lambert et al. (2022), de Bodt et al. (2025), and Kury et al. (2025)). Note that Lambert et al. (2022) converts the MDS loss to quartet stress (a loss over four samples) and relative distance (distance divided by the sum of six quartet distances), enabling global regularization and lower computation than majorization.

H Construction of the High-Dimensional Graph

Stochastic Neighbor Embedding (SNE) Hinton and Roweis (2002) underpins modern dimensionality reduction algorithms. It constructs a high-dimensional graph for $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ through

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_i - x_t\|_2^2 / 2\sigma_i^2)}, \quad (87)$$

where σ_i^2 is chosen to match the user-defined perplexity P , defined as

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

UMAP instead constructs its high-dimensional graph from the k-nearest-neighbor (k-NN) graph:

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ is the set of k nearest neighbors of x_i , and σ_i is chosen so that $\sum_j p_{i|j} = \log_2(k)$. UMAP then symmetrizes the graph with a t-conorm:

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP uses the k-NN graph for affinities, all set to 1, together with a self-tuning distance measure (Zelnik-Manor and Perona [2004]),

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the average distance between x_i and its fourth through sixth Euclidean nearest neighbors. This parameter serves the same purpose as the corresponding σ_i parameters in Eq. (87) and (90), even though the definitions differ.

Regardless of these graph-construction choices, the subsequent optimization is governed primarily by the attraction and repulsion shapes identified in our analysis.

I Detailed Results for Varying λ_a and λ_b

This section provides the detailed results behind Fig. 11(h,i). We obtain these results by replacing UMAP’s attraction and repulsion shapes with the corresponding shapes from other methods. Figure 18 reproduces the main-text results. Figures 19-22 show the individual embeddings for each choice of λ_a and λ_r for each method, together with the initialization, an annealed-learning-rate reference embedding, and the ablations where either λ_a or λ_r is set to 0. For PaCMAP, which uses mid-near pairs, we also include a reference with mid-near pairs enabled. Figures 23-27 and Figures 28-32 extend the same evidence to FMNIST and single-cell transcriptomes, respectively.

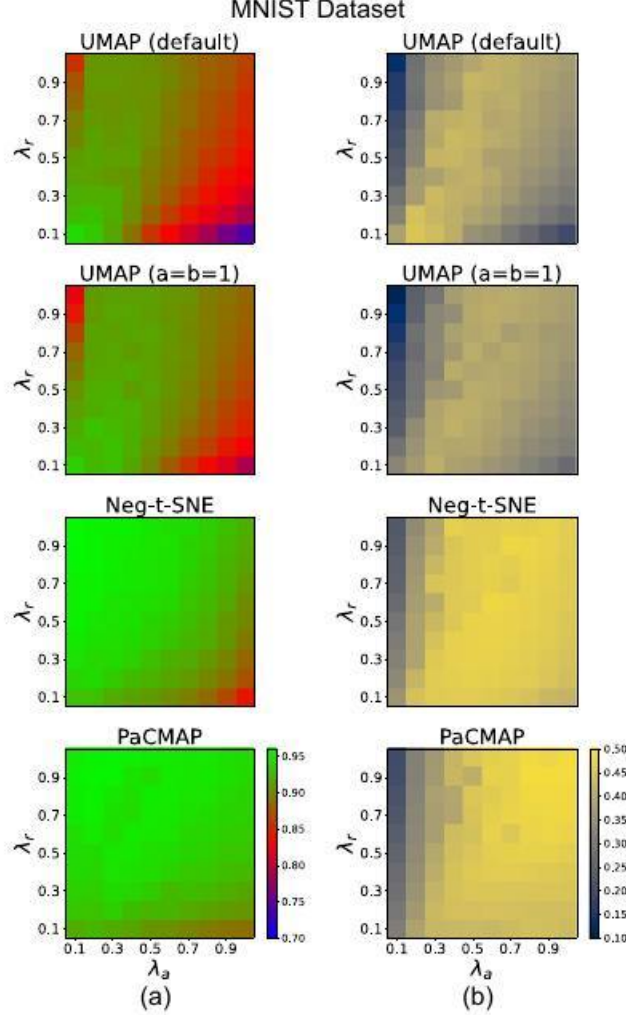


Figure 18: MNIST confirms the shape-level tradeoff across methods. (a) Trustworthiness and (b) Silhouette score for different methods.

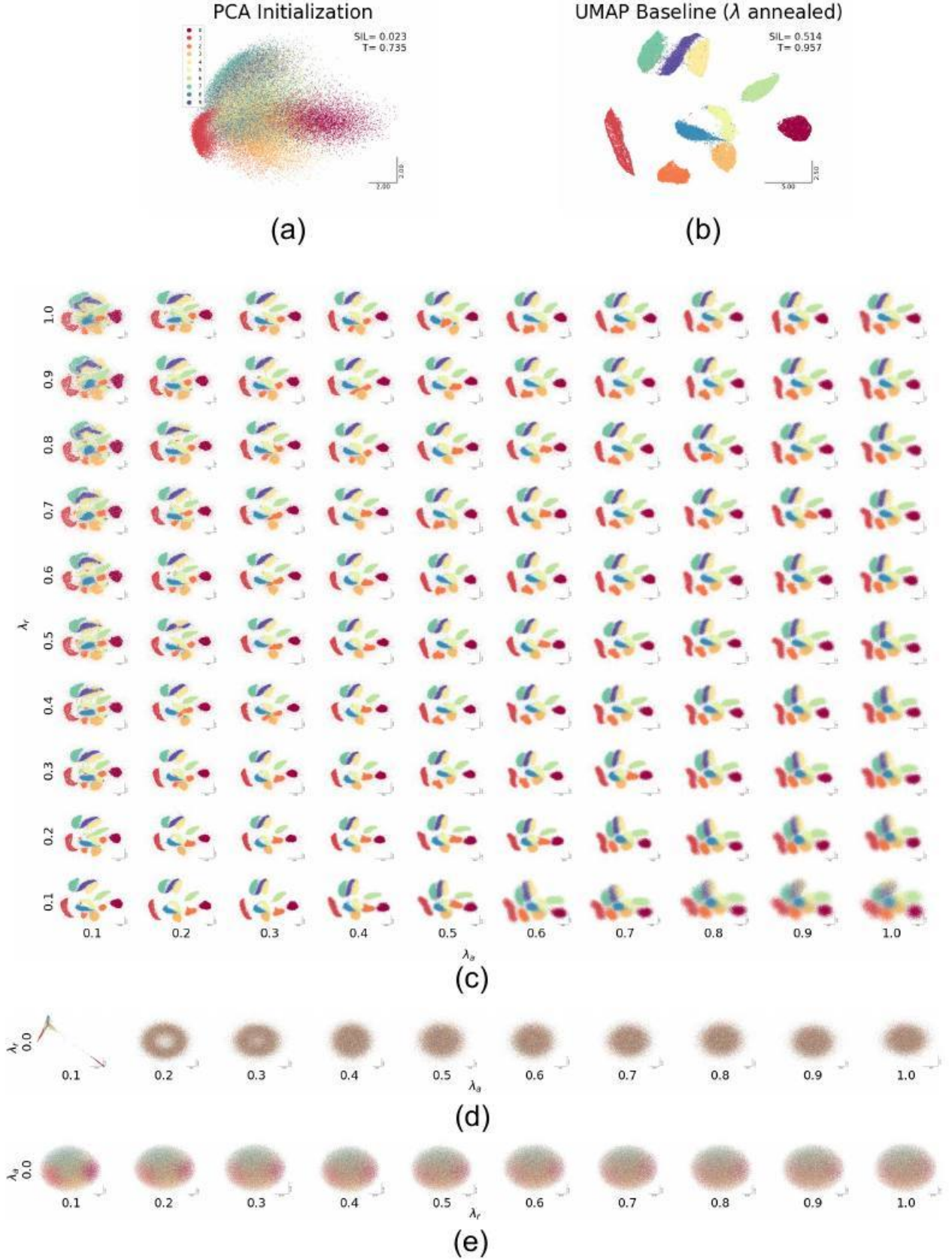


Figure 19: Varying λ_a and λ_r on MNIST with UMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Annealed baseline with λ initialized at 1. (c) Embeddings for fixed λ_a and λ_r , without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone does not produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone does not produce clusters.

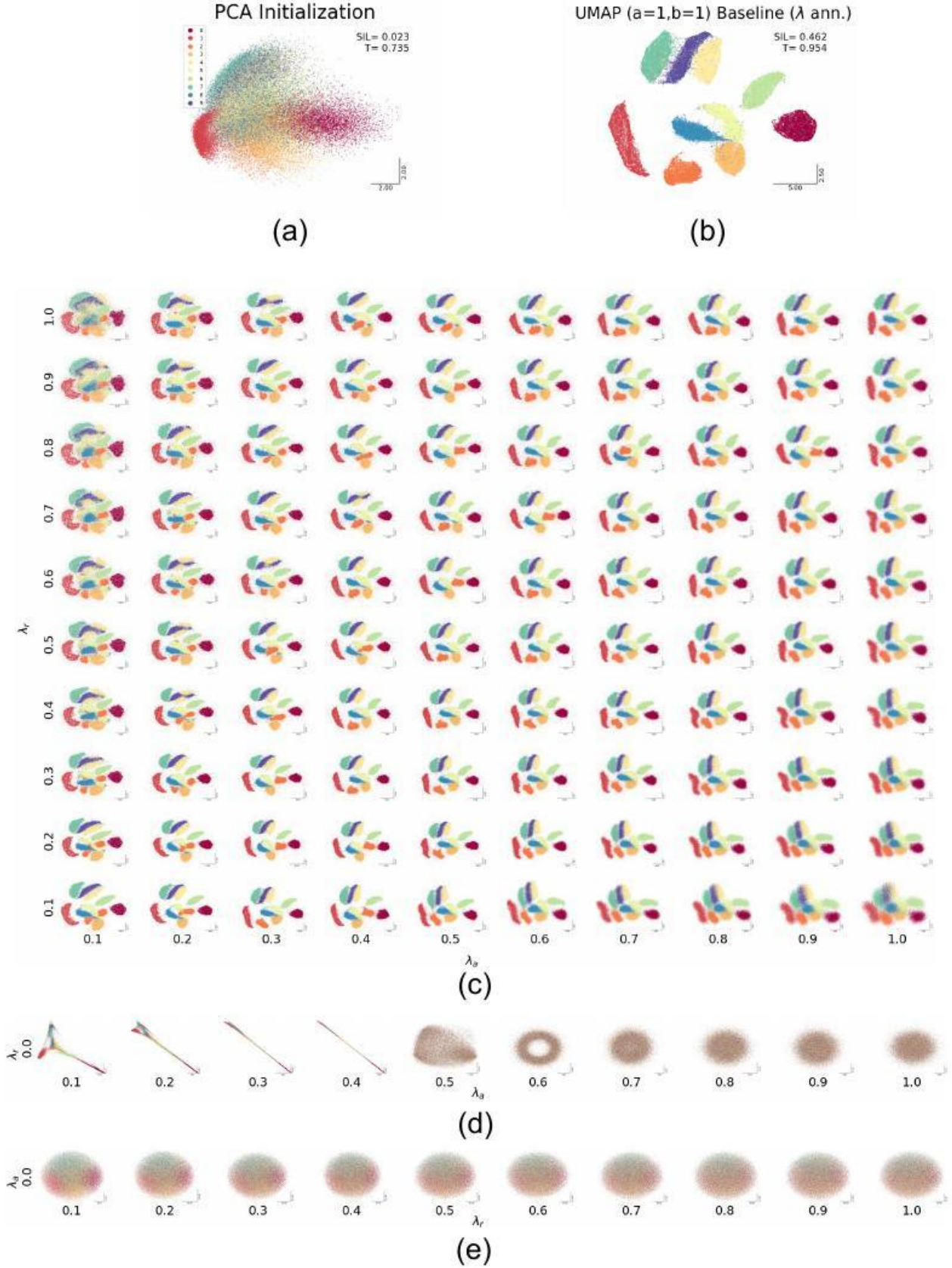


Figure 20: Varying λ_a and λ_r on MNIST with UMAP shapes at $a = 1$ and $b = 1$. (a) Embedding initialization. (b) Annealed baseline with λ initialized at 1. (c) Embeddings for fixed λ_a and λ_r without annealing. (d) With $\lambda_r = 0$, attraction alone does not produce clusters. (e) With $\lambda_a = 0$, repulsion alone does not produce clusters.

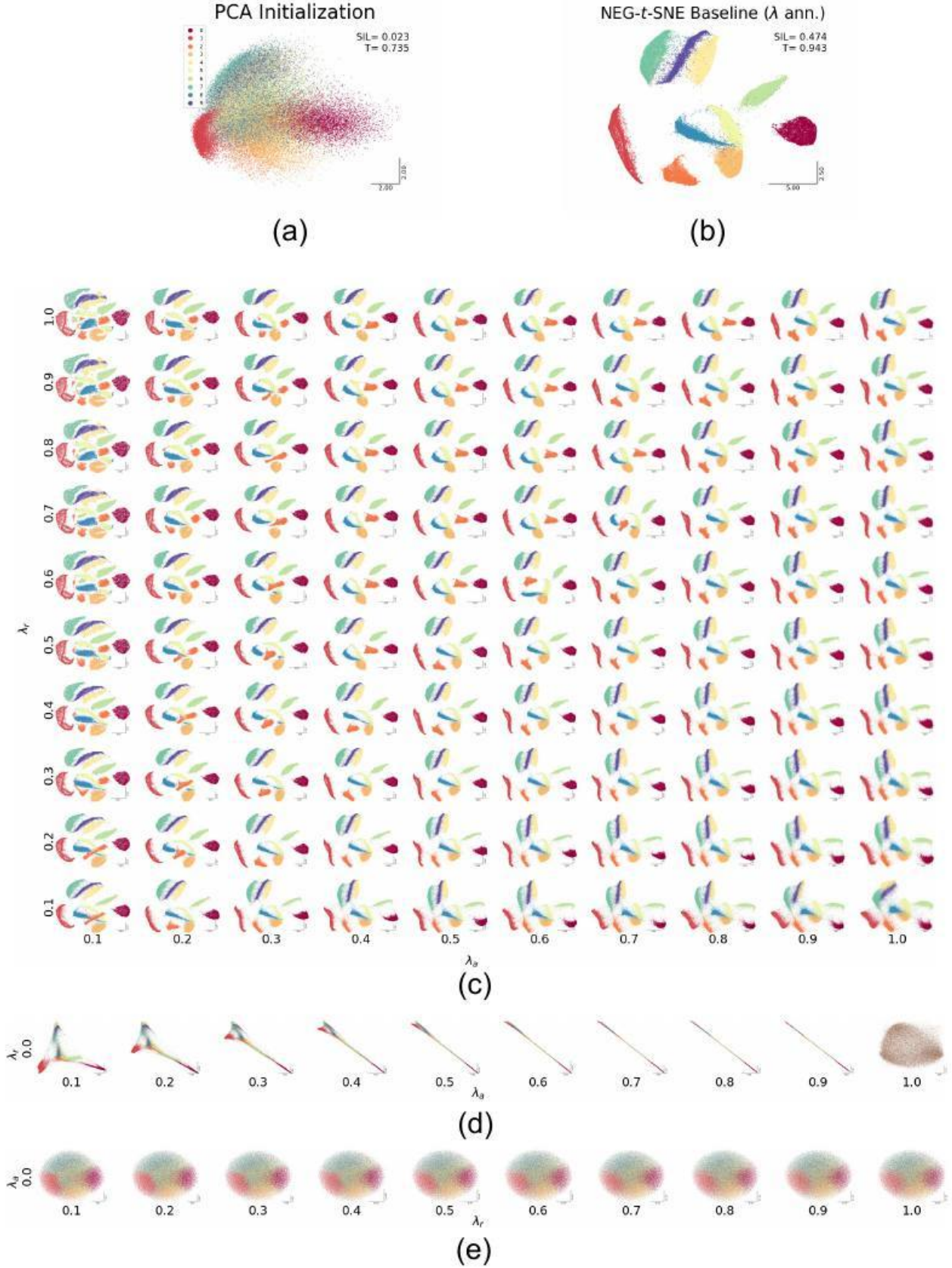


Figure 21: Varying λ_a and λ_r on MNIST with NEG-t-SNE attraction and repulsion shapes. (a) Embedding initialization. (b) Annealed baseline with λ initialized at 1. (c) Embeddings for fixed λ_a and λ_r without annealing. (d) With $\lambda_r = 0$, attraction alone does not produce clusters. (e) With $\lambda_a = 0$, repulsion alone does not produce clusters.

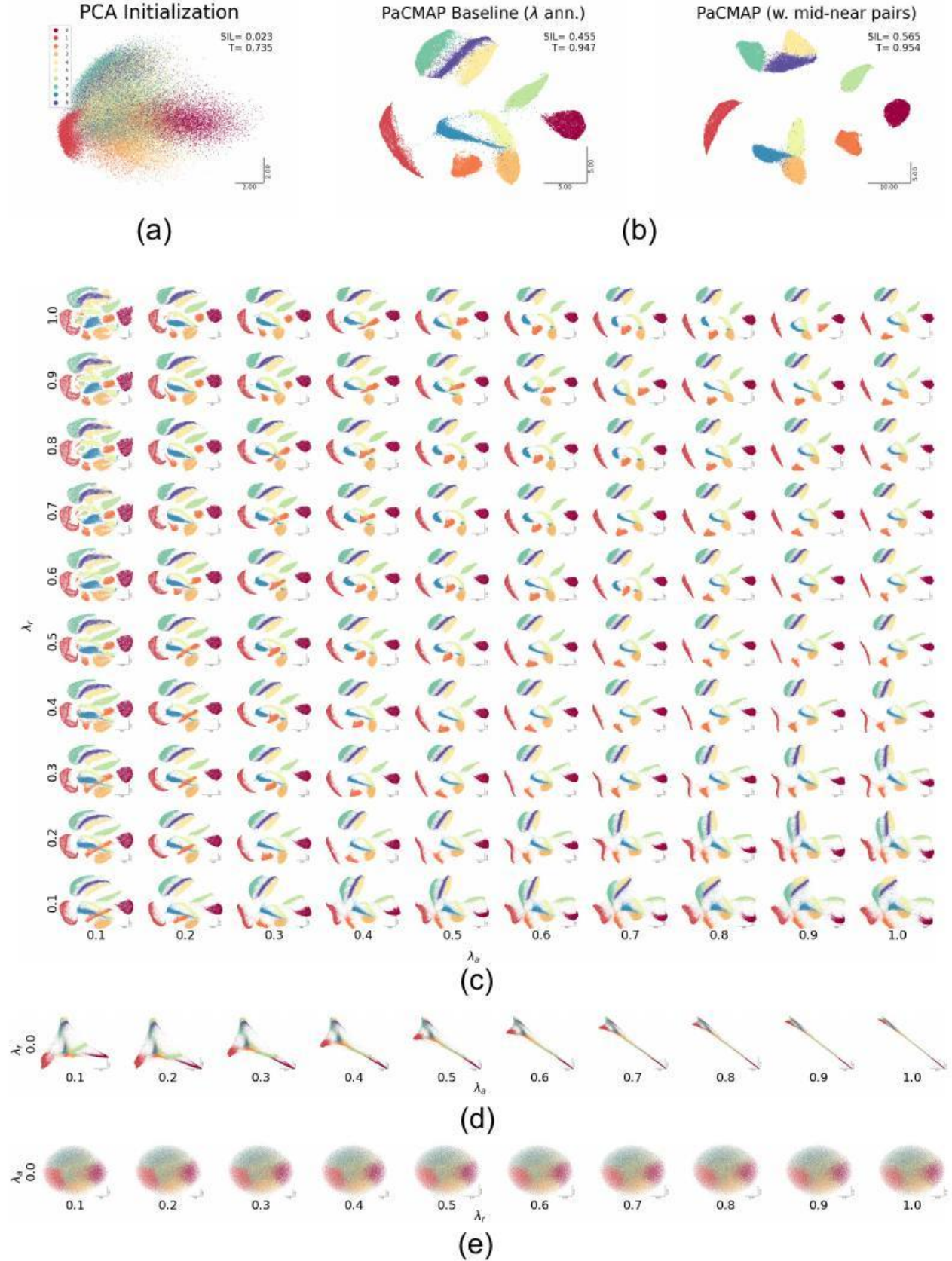


Figure 22: Varying λ_a and λ_r on MNIST with PaCMAP attraction and repulsion shapes. (a) Embedding initialization. (b) Left: annealed baseline with λ initialized at 1 and mid-near points excluded to isolate attraction-repulsion; right: reference with mid-near points included. (c) Embeddings for fixed λ_a and λ_r without annealing. (d) With $\lambda_r = 0$, attraction alone does not produce clusters. (e) With $\lambda_a = 0$, repulsion alone does not produce clusters.

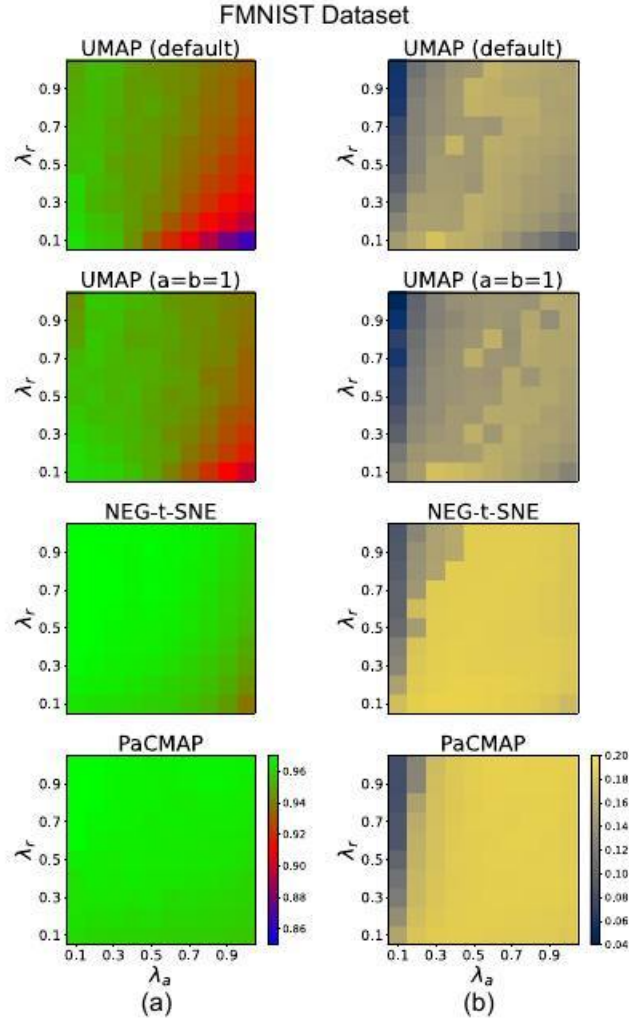


Figure 23: FMNIST confirms the shape-level tradeoff across methods. (a) Trustworthiness and (b) Silhouette score for different methods.

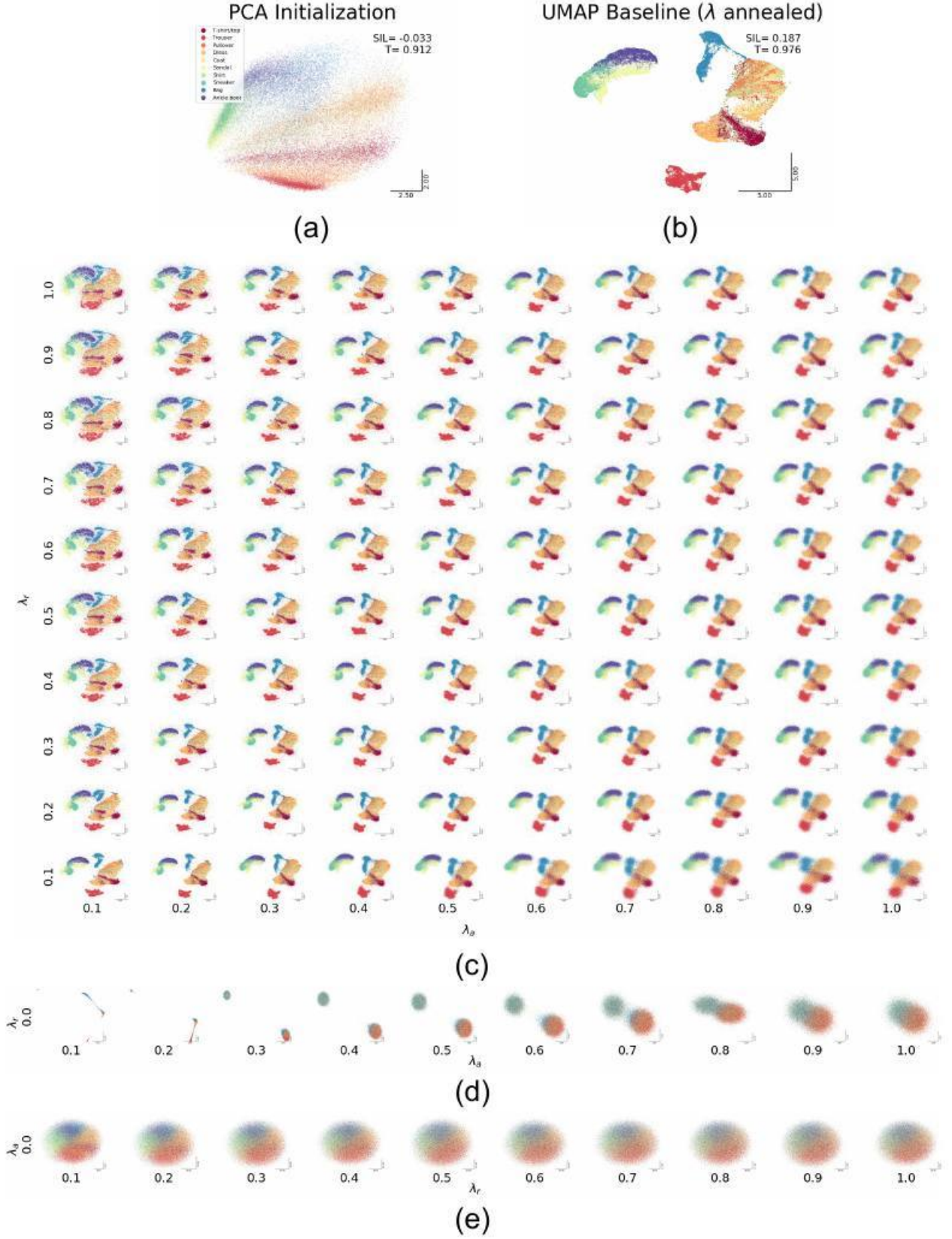


Figure 24: Varying λ_a and λ_r on FMNIST with UMAP attraction and repulsion shapes. (a) Embedding initialization. (b) Annealed baseline with λ initialized at 1. (c) Embeddings for fixed λ_a and λ_r without annealing. (d) With $\lambda_r = 0$, attraction alone does not produce clusters. (e) With $\lambda_a = 0$, repulsion alone does not produce clusters.

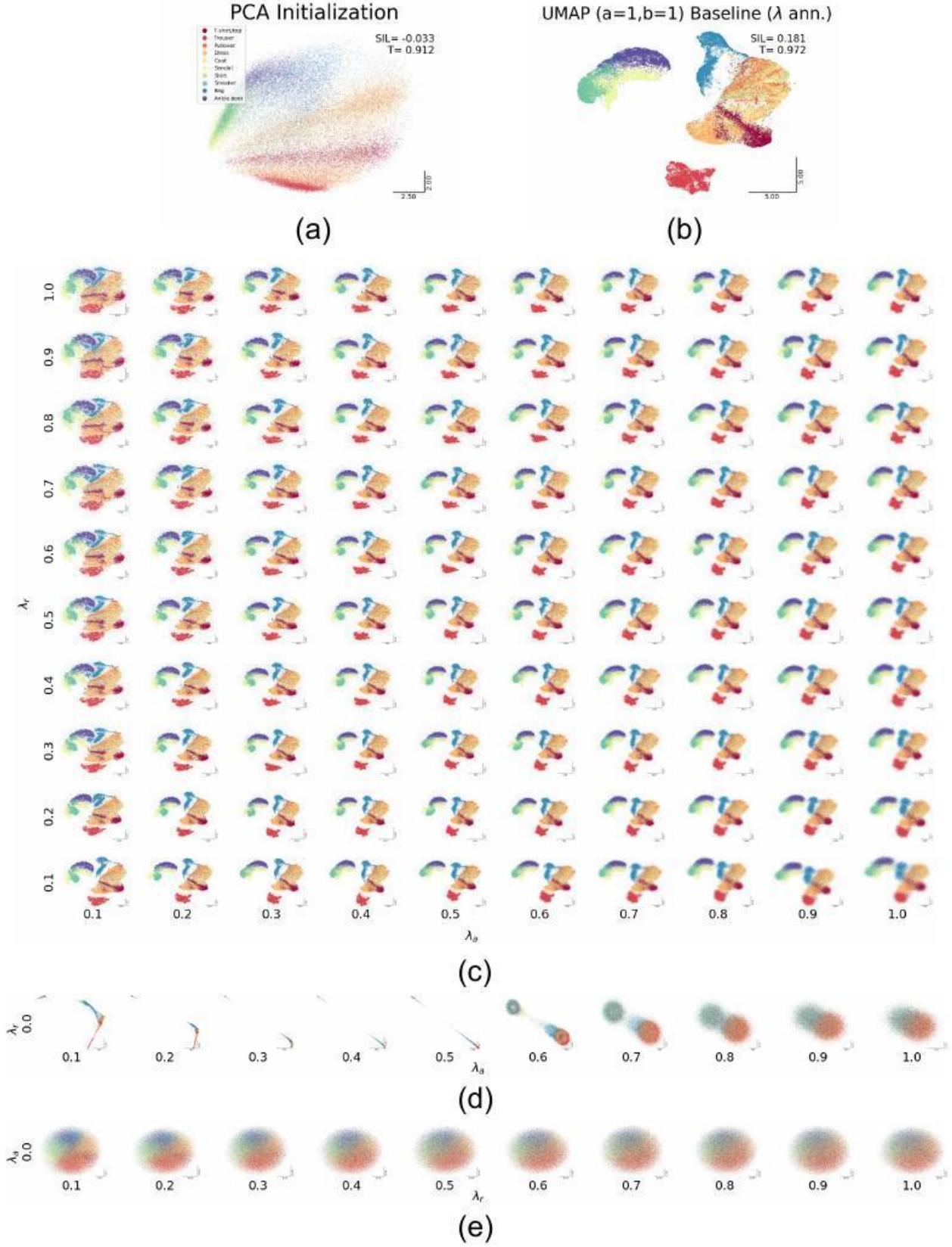


Figure 25: Varying λ_a and λ_r on FMNIST with UMAP shapes at $a = 1$ and $b = 1$. (a) Embedding initialization. (b) Annealed baseline with λ initialized at 1. (c) Embeddings for fixed λ_a and λ_r without annealing. (d) With $\lambda_r = 0$, attraction alone does not produce clusters. (e) With $\lambda_a = 0$, repulsion alone does not produce clusters.

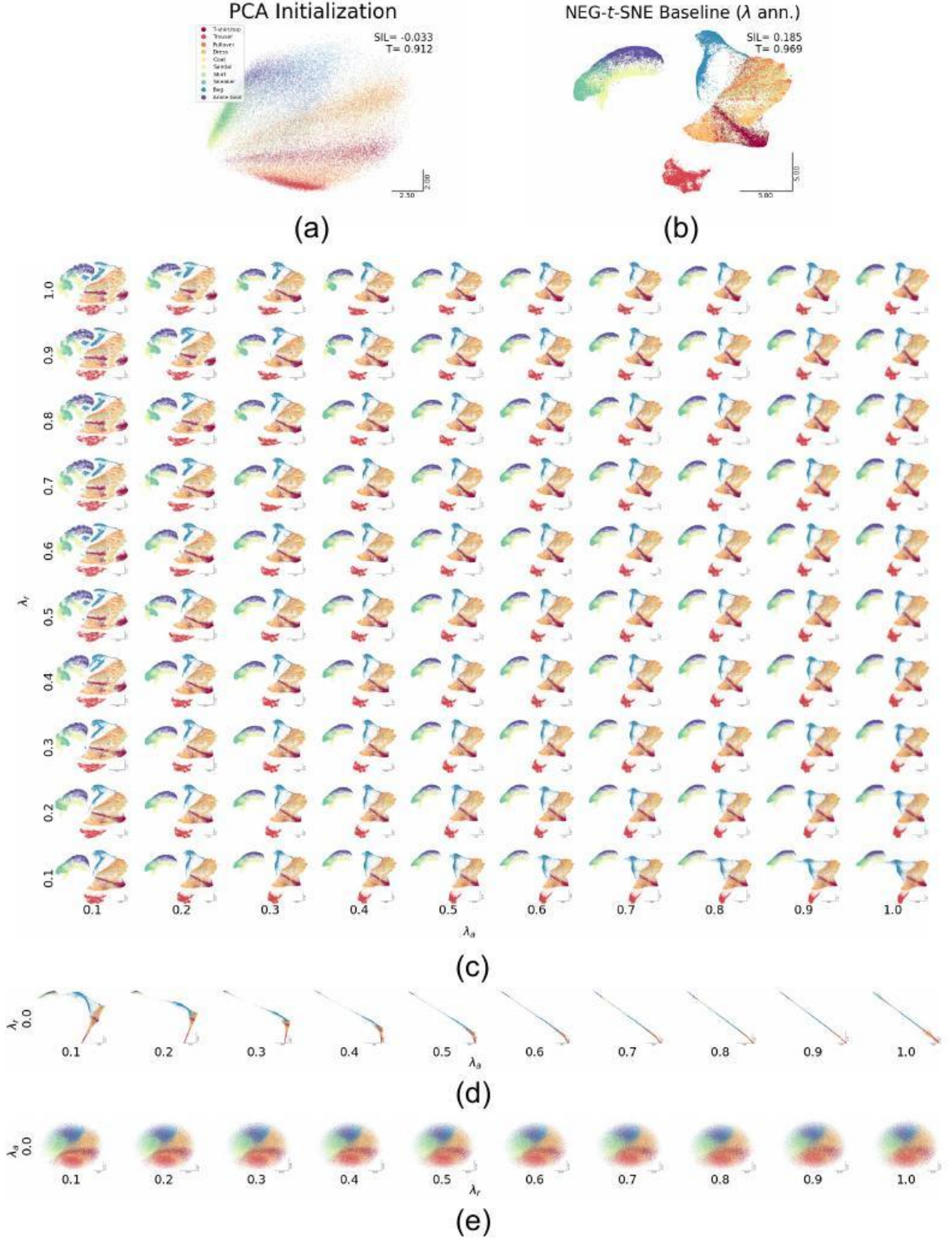


Figure 26: Varying λ_a and λ_r on FMNIST with NEG-t-SNE attraction and repulsion shapes. (a) Embedding initialization. (b) Annealed baseline with λ initialized at 1. (c) Embeddings for fixed λ_a and λ_r without annealing. (d) With $\lambda_r = 0$, attraction alone does not produce clusters. (e) With $\lambda_a = 0$, repulsion alone does not produce clusters.

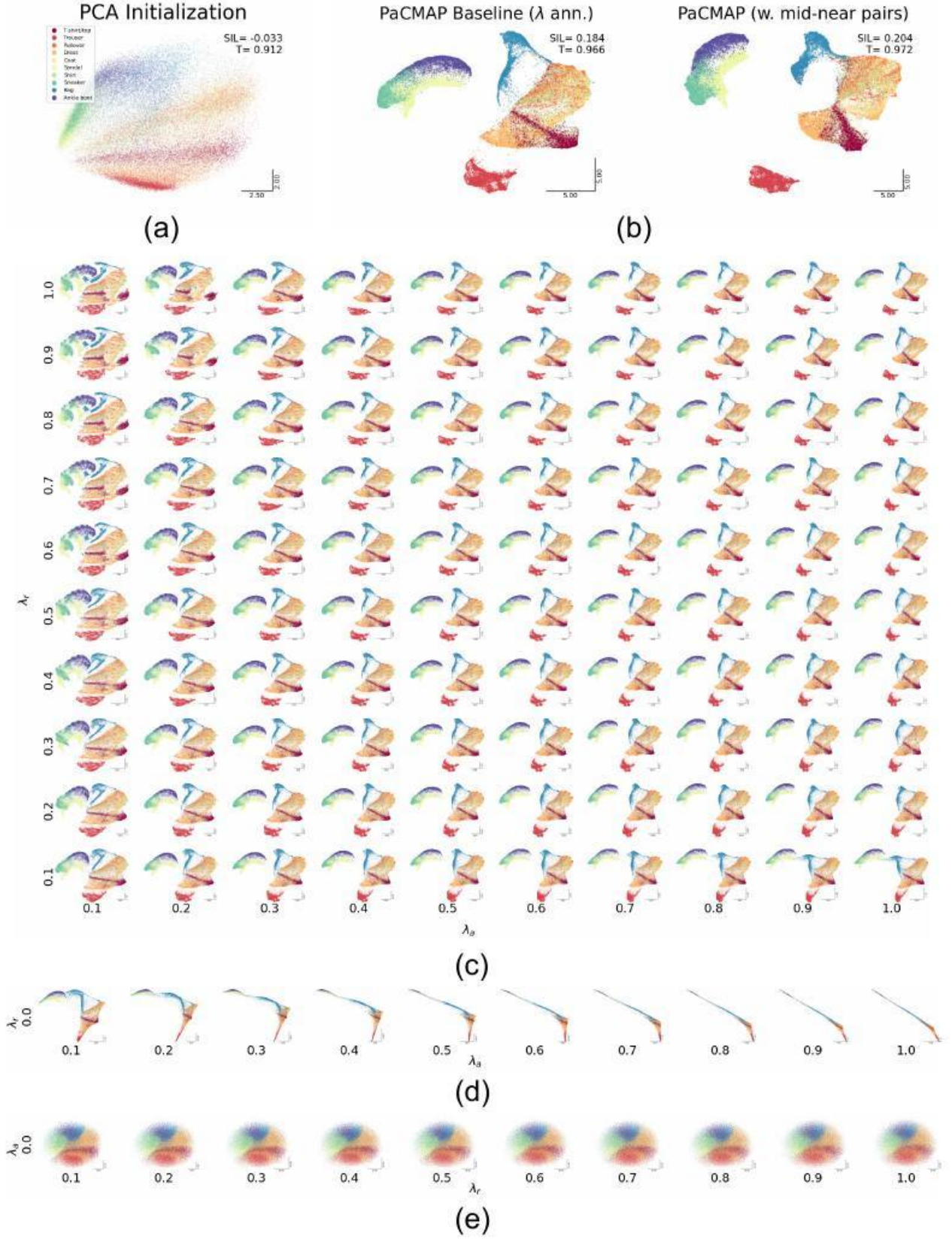


Figure 27: Varying λ_a and λ_r on FMNIST with PaCMAP attraction and repulsion shapes. (a) Embedding initialization. (b) Left: annealed baseline with λ initialized at 1 and mid-near points excluded to isolate attraction-repulsion; right: reference with mid-near points included. (c) Embeddings for fixed λ_a and λ_r without annealing. (d) With $\lambda_r = 0$, attraction alone does not produce clusters. (e) With $\lambda_a = 0$, repulsion alone does not produce clusters.

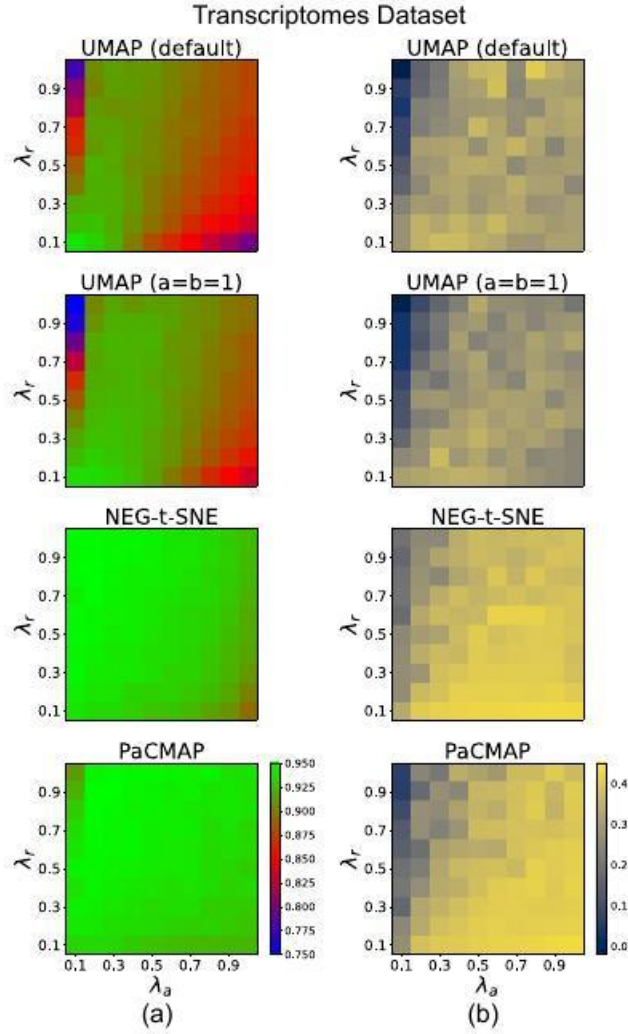


Figure 28: Single-cell transcriptomes confirm the shape-level tradeoff across methods. (a) Trustworthiness and (b) Silhouette score for different methods.

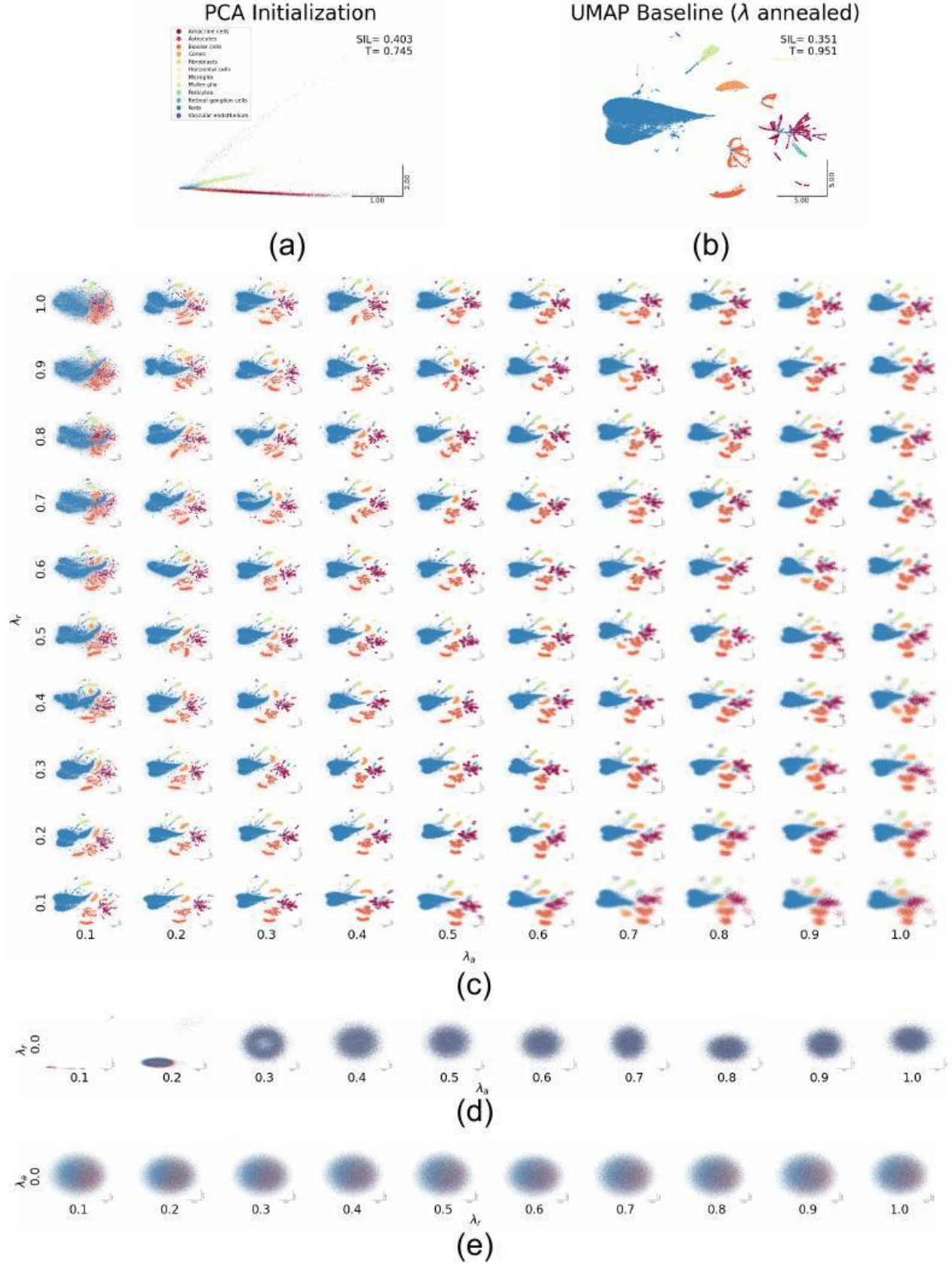


Figure 29: Varying λ_a and λ_r on single-cell transcriptomes with UMAP attraction and repulsion shapes. (a) Embedding initialization. (b) Annealed baseline with λ initialized at 1. (c) Embeddings for fixed λ_a and λ_r without annealing. (d) With $\lambda_r = 0$, attraction alone does not produce clusters. (e) With $\lambda_a = 0$, repulsion alone does not produce clusters.

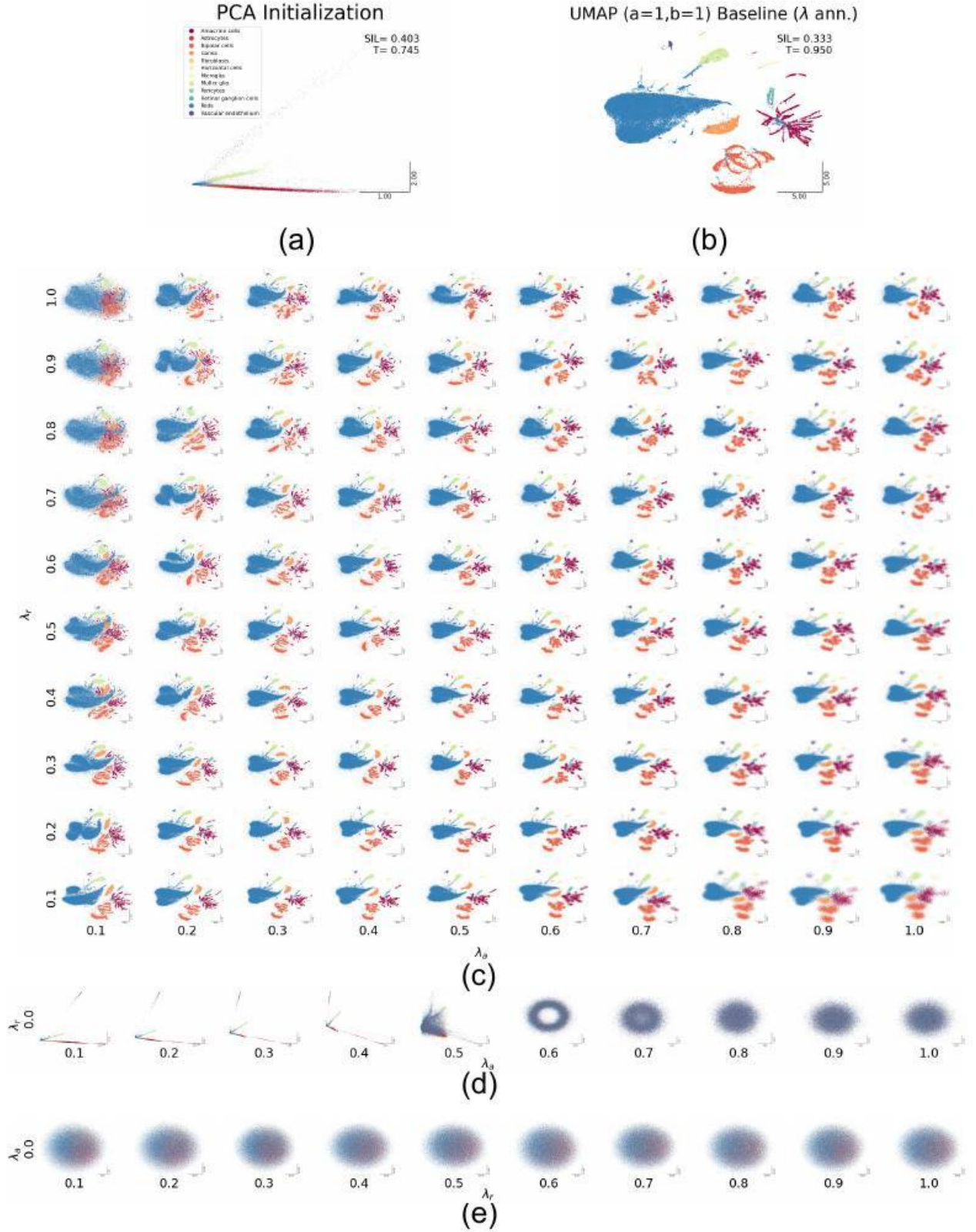


Figure 30: Varying λ_a and λ_r on single-cell transcriptomes with UMAP shapes at $a = 1$ and $b = 1$. (a) Embedding initialization. (b) Annealed baseline with λ initialized at 1. (c) Embeddings for fixed λ_a and λ_r without annealing. (d) With $\lambda_r = 0$, attraction alone does not produce clusters. (e) With $\lambda_a = 0$, repulsion alone does not produce clusters.

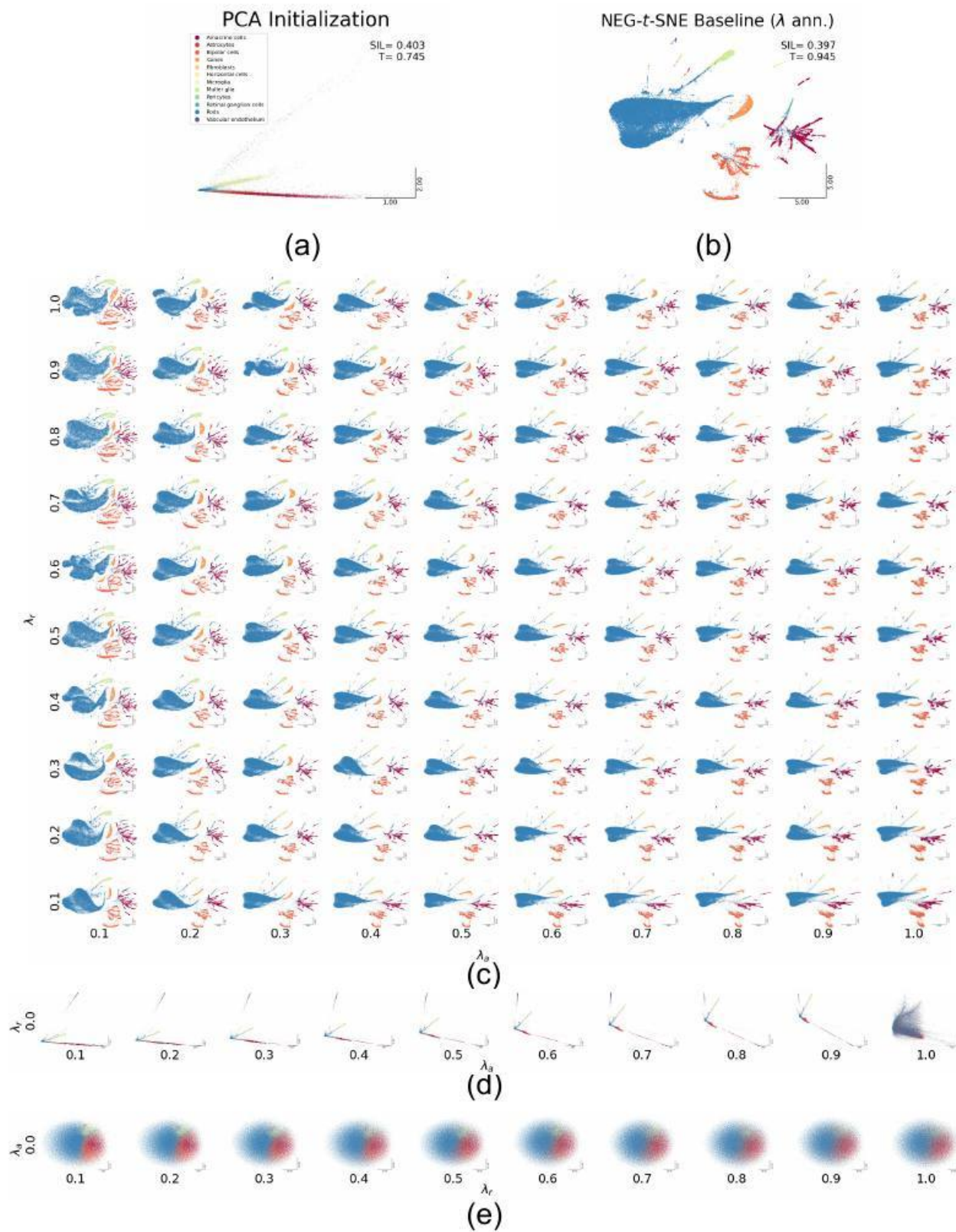


Figure 31: Varying λ_a and λ_r on single-cell transcriptomes with NEG-*t*-SNE attraction and repulsion shapes. (a) Embedding initialization. (b) Annealed baseline with λ initialized at 1. (c) Embeddings for fixed λ_a and λ_r without annealing. (d) With $\lambda_r = 0$, attraction alone does not produce distinct clusters. (e) With $\lambda_a = 0$, repulsion alone does not produce clusters.

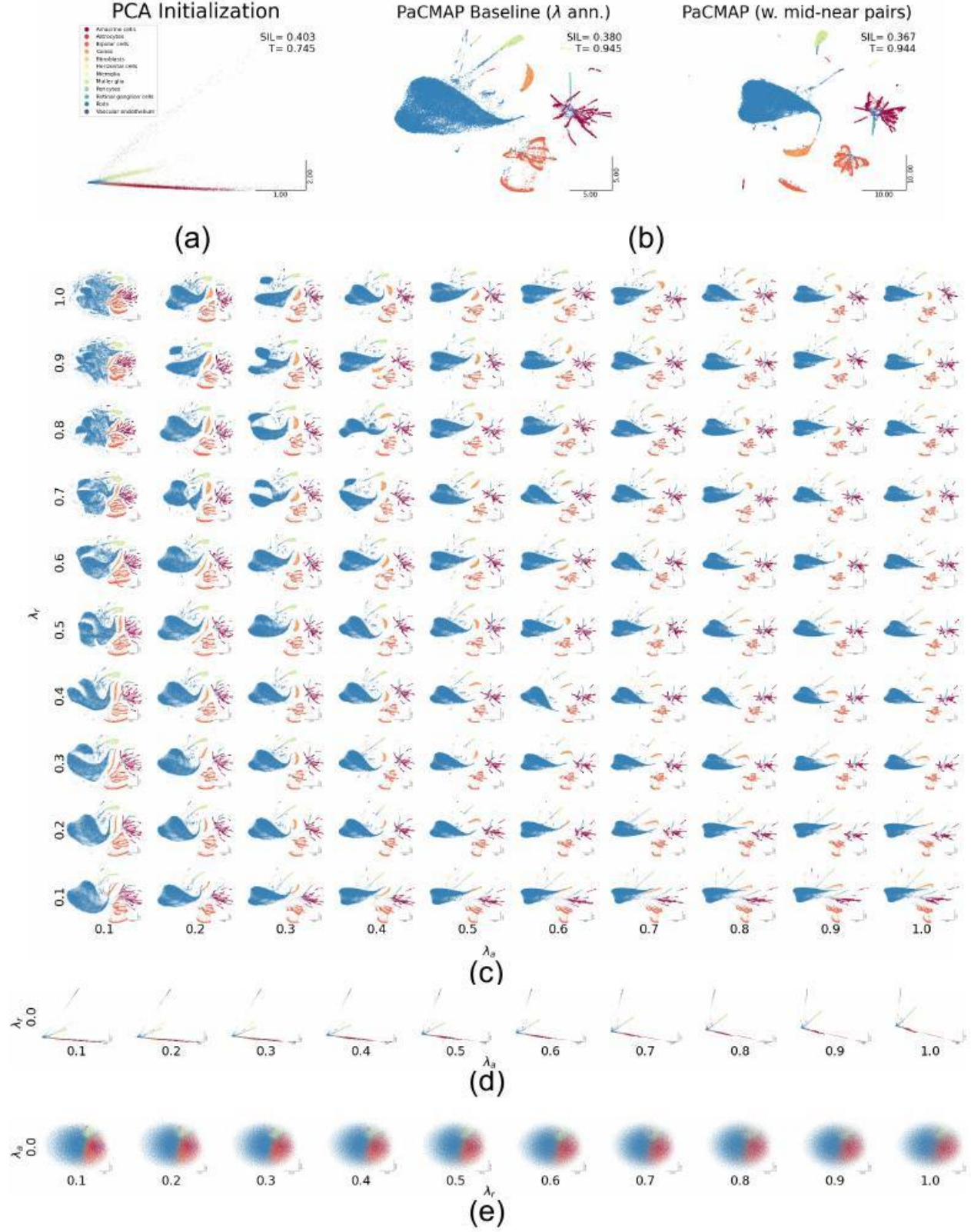


Figure 32: Varying λ_a and λ_r on single-cell transcriptomes with PaCMAP attraction and repulsion shapes. (a) Embedding initialization. (b) Left: annealed baseline with λ initialized at 1 and mid-near points excluded to isolate attraction-repulsion; right: reference with mid-near points included. (c) Embeddings for fixed λ_a and λ_r without annealing. (d) With $\lambda_r = 0$, attraction alone does not produce clusters. (e) With $\lambda_a = 0$, repulsion alone does not produce clusters.

J Mix and Match Approach

Our analysis treats attraction and repulsion shapes as independently meaningful design components. This immediately suggests a mix-and-match strategy: combine attraction and repulsion shapes derived from different losses and principles. Figure 33 demonstrates this strategy. We restrict attraction shapes to $[-1, 0]$ so that Proposition 4.1 applies and the embeddings become less learning-rate sensitive. We also use finite repulsion shapes to avoid excessively large short-distance repulsive forces. The learning rate is held constant ($\lambda = 1$), and all variants use the same UMAP optimization, so the observed changes are attributable to the shapes rather than to learning-rate scheduling or optimizer differences. Overall, UMAP’s attraction shape (via $f_a = 0.5f_a^U$) gives stronger clustering, reflected in higher silhouette scores, whereas PaCMAP’s attraction shape gives better local structure, reflected in higher trustworthiness. PaCMAP’s advantage in local structure follows from its attraction shape remaining within $[-0.5, 0]$, which prevents flips during updates. Confining UMAP and NEG- t -SNE attraction shapes to $[-0.5, 0]$ (via $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$, respectively) improves local structure at the cost of clustering performance (Fig. 34).

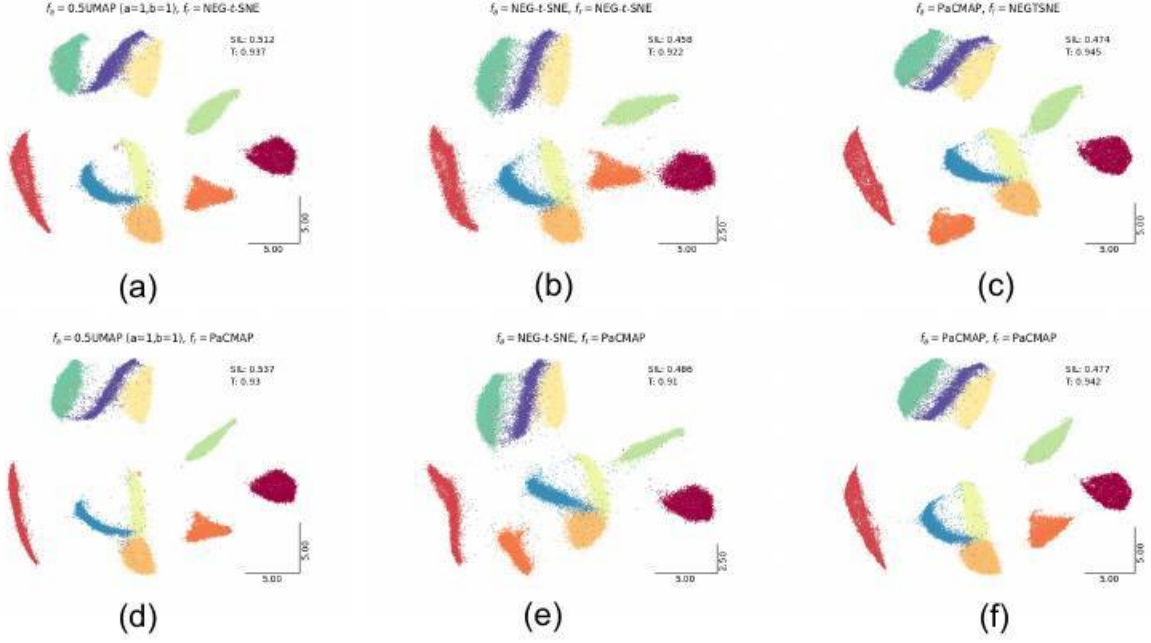


Figure 33: MNIST embeddings from mixing attraction and repulsion shapes at constant learning rate ($\lambda = 1$). Attraction shapes satisfy Proposition 4.1 and repulsion shapes are finite. Top row: UMAP, NEG- t -SNE, and PaCMAP attraction shapes paired with NEG- t -SNE repulsion. Bottom row: UMAP, NEG- t -SNE, and PaCMAP attraction shapes paired with PaCMAP repulsion.

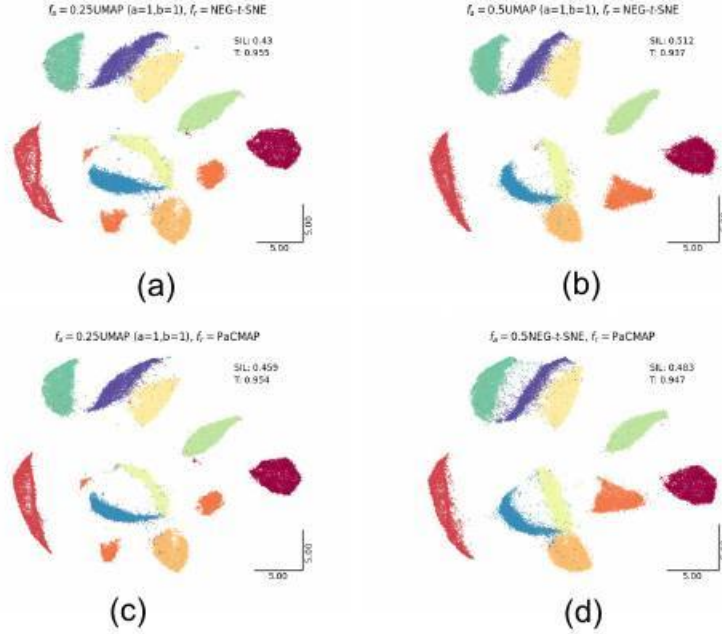


Figure 34: MNIST embeddings from mix-and-match shapes with attraction confined to $[-0.5, 0]$ at constant learning rate ($\lambda = 1$). This restriction prevents flips during attractive updates. Top row: UMAP and NEG- t -SNE attraction shapes paired with NEG- t -SNE repulsion. Bottom row: UMAP and NEG- t -SNE attraction shapes paired with PaCMAP repulsion.

K Implementation Details

For the analysis, we implemented UMAP directly so that attraction and repulsion shapes could be controlled explicitly. We used `numba` (Lam et al. 2015) to compute an exact nearest-neighbor graph, rather than an approximate one, with $k = 15$, and used `scikit-learn`'s (Pedregosa et al. 2011) PCA implementation for PCA initialization. This implementation produced and quantified the embeddings in Figs. 1, 2, 3, 4, 6, 8, 9, and 10. We used the same implementation when replacing UMAP's attraction and repulsion shapes with those from alternative methods (Figs. 18, 34 unless otherwise stated). Trustworthiness and silhouette scores were computed with the corresponding `scikit-learn` functions.

The mappings in Figs. 2, 3, 9, 10, and 15 are rotated to their respective reference embeddings (panel (a) in each figure). We performed this alignment by normalizing embeddings to zero mean and unit norm and then using SciPy's (Virtanen et al. 2020) `orthogonal_procrustes` method to extract rotation and scaling parameters.

For the Neg- t -SNE comparison in Fig. 11 we used the original contrastive-embedding framework implementation for both UMAP and Neg- t -SNE; the implementation details will be released upon acceptance.

PaCMAP and LocalMAP embeddings in Figs. 15, 16, 22 (b) (right), 27 (b) (right), and 32 (b) (right) were generated with the official PaCMAP package; the implementation details will be released upon acceptance.

The code for displaying the PubMed embeddings in Fig. 7 will be released upon acceptance.